

When Markets Never Sleep: Intraday Liquidity Patterns and Volatility Effects in Cryptocurrency Trading

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Abstract

This paper examines the microstructure of cryptocurrency markets by analyzing liquidity and volatility patterns. Using high-frequency 15-minute data from Binance and Coinbase, we study Bitcoin and Ethereum trading pairs with both fiat currencies and stablecoins. We employ a depth-adjusted spread measure standardized for trade size to identify and statistically validate daily and intraday periodicities in spreads and volatility across trading venues and quote currencies. Our findings reveal that volatility is the primary driver of spread variation, though periodic factors provide additional explanatory power in an integrated model. The identified periodicities enhance out-of-sample spread forecasting performance across several markets, with consistent differences observed between venues and quote currencies.

Keywords: cryptocurrency, liquidity, spread, volatility, intraday, forecasting.

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Abstract

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Keywords: Cryptocurrency, Market Microstructure, Liquidity, Volatility, High-frequency data

JEL codes: G12, G14, G15, C58

1 Introduction

Liquidity is a first-order input into trading costs and price discovery, while volatility fundamentally shapes market dynamics and risk assessment. In market microstructure, the bid–ask spread compensates liquidity suppliers for order-processing costs, inventory risk, and adverse selection, and therefore scales with return variance and the probability of informed order flow (Amihud and Mendelson, 1980; Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997; Stoll, 1989). Traditional markets display pronounced intraday regularities in volatility, trading activity, and spreads that align with information arrival (Admati and Pfleiderer, 1988; Bessembinder, 1994; French and Roll, 1986; Mcinish and Wood, 1992). Cryptocurrencies add two features that make both liquidity measurement and volatility modeling particularly challenging: round-the-clock trading on fragmented venues, and frequent shifts in participation between retail and institutional flow (Brauneis et al., 2021; Makarov and Schoar, 2020). Understanding how these forces map into both liquidity costs and volatility patterns in crypto is the focus of this paper.

To address these issues, we collect high-frequency 15-minute limit-order-book data for Bitcoin and Ethereum on two major centralised exchanges, Binance and Coinbase, each traded against fiat currencies (USD, EUR) and stablecoins (USDT, USDC, FDUSD). We construct a depth-adjusted volume-weighted percent quoted spread for a given trade size of USD 100K to measure tightness across venues and quote currencies in a decision-relevant way, complementing inside-quote proxies common in the literature (Brauneis et al., 2021; Fong et al., 2017). We then examine the intraday patterns of these spreads and their synchronization with volatility across assets, venues, and quote currencies, focusing on the timing relative to Asian, European and U.S. trading sessions. This builds on recent evidence that volatility and activity in crypto display strong time-of-day seasonality centered on European–U.S. hours (Baur et al., 2019; Eross et al., 2019; Hansen et al., 2024a; Wang et al., 2020). To characterize these intraday liquidity cycles, we apply the Fast Fourier Transform (FFT) and harmonic analysis to identify cyclical components and dominant frequencies in

spread time series, testing whether observed periodicity represents a genuine feature rather than an artefact of averaging (Andersen and Bollerslev, 1997). Following the inventory- and information-risk approaches (Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997), we estimate an integrated model that incorporates volatility measures (both EWMA and instantaneous r_t^2), contemporaneous and lagged returns, and periodic factors to examine the relative importance of each variable in explaining spread dynamics. Finally, we test whether explicitly incorporating periodicity improves out-of-sample spread forecasting performance across multiple currency pairs.

This paper contributes to the cryptocurrency microstructure literature in four ways. First, we provide high-frequency, market-wide evidence in crypto that depth-adjusted, volume-weighted percent quoted spread computed at a fixed notional scales with volatility and information intensity, consistent with classic inventory and adverse-selection frameworks (Amihud and Mendelson, 1980; Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997; Stoll, 1989). Second, we extend intraday results from equities and FX (Admati and Pfleiderer, 1988; Bessembinder, 1994; French and Roll, 1986; McInish and Wood, 1990; McInish and Wood, 1992) and recent crypto work (Baur et al., 2019; Eross et al., 2019; Hansen et al., 2024a; Wang et al., 2020) by showing that spreads themselves — measured at fixed notional of USD 100k — exhibit strong daily periodicity with synchronized phases across assets, namely Bitcoin and Ethereum, paired with both fiat currencies and stablecoins, and venues such as Coinbase and Binance. We document and statistically validate periodicity in spreads — not only in volatility — that peaks during U.S. trading hours on weekdays and flattens on weekends. Third, our depth-adjusted spread complements widely used low-frequency proxies (Fong et al., 2017) and aligns with recent exchange-level evidence on the determinants of liquidity and fee-policy effects (Brauneis et al., 2022a,b; Galati, 2024). The cross-venue and quote-currency differences that we document contribute additional evidence on market fragmentation and incomplete integration (Makarov and Schoar, 2020). Lastly, by embedding a periodic factor in a forecasting model, we show measurable gains for some

pairs, echoing the importance of deterministic seasonals in high-frequency econometrics (Andersen and Bollerslev, 1997) and clarifying when such gains are economically meaningful for market makers and algorithmic execution.

2 Microstructure foundations

In dealership markets, the bid–ask spread compensates dealers for the provision of immediacy and the costs of carrying inventory, so spreads embed a return for supplying liquidity under risk (Amihud and Mendelson, 1980). Cross-sectional evidence from over-the-counter trading shows that spreads rise with proxies for order-processing costs, risk, and price level (Benston and Hagerman, 1974). The components view—order processing, inventory, and adverse selection—is then articulated and taken to the data in structural and reduced-form decompositions (Huang and Stoll, 1997; Stoll, 1989).

On the information side, Copeland and Galai (1983) interpret the spread as compensation for the option value of trading against better-informed counterparties, predicting wider spreads when information asymmetry intensifies. In a dynamic setting, Kyle (1985) show that an informed trader optimally feeds order flow into the book; the price-impact coefficient (λ) increases when depth is scarce or information asymmetry is high, implying higher execution costs and, by extension, wider effective spreads. Information-based models imply that dealers protect themselves by widening quotes when the probability of informed order flow is higher (Glosten and Milgrom, 1985). Also, Easley and O’Hara (1987) relate trade size and price changes to the probability of informed trading, yielding testable predictions that spreads and price impact are larger when informed-trade likelihood rises. These mechanisms are fully consistent with specialist/limit-order markets in which dealers and liquidity suppliers manage both inventory risk and adverse-selection risk in real time (Glosten and Milgrom, 1985; Ho and Stoll, 1981).

In equities, intraday spread dynamics track precisely the risk-and-information logic of

dealership and adverse-selection models. Spreads are reliably wider at information-rich times—around the open and close—and co-move positively with contemporaneous volatility and order-flow imbalances, consistent with dealers managing inventory and protecting against informed order flow (Hasbrouck, 1991; Mcinish and Wood, 1992)¹. Structural decompositions attribute a sizable share of the effective spread to adverse selection and inventory components, which rise when return variance is high (Huang and Stoll, 1997; Stoll, 1989). Periods of heightened market-wide activity further amplify price impact and reduce depth, generating commonality in liquidity across names (Hasbrouck, 2001). The time-aggregation evidence points in the same direction: return variance is markedly higher during trading than non-trading intervals, implying that the intraday schedule concentrates information arrivals, risk, and transaction costs (French and Roll, 1986). Theoretical predictions of strategic crowding of order flow at focal times—when both informed and uninformed traders prefer to transact—help rationalize why spreads can remain elevated even when quoted competition is intense (Admati and Pfleiderer, 1988).

In interbank FX, quoted and effective spreads widen with proxies for price risk and ahead of non-trading intervals, exactly as inventory-risk and information-risk models predict (Bessembinder, 1994)². More generally, microstructure models and empirical evidence show that higher volatility raises the cost of immediacy, so the price-impact coefficient and the spread move together (Copeland and Galai, 1983; Kyle, 1985). Cross-market studies document that liquidity conditions co-vary across asset classes as shared risk and funding conditions shift, again indicating a risk-based core to the cost of immediacy (Chordia et al., 2005)³.

A parallel empirical literature has begun to trace the determinants of trading costs in

¹Large trades cause the spread to widen; trades occurring in the face of wide spreads have larger price impacts; and, information asymmetries are more significant for smaller firms (Hasbrouck, 1991).

²High-frequency announcement work shows that spreads and volatility spike around scheduled U.S. macro releases, with sharp, short-lived deteriorations in liquidity as dealers face greater adverse-selection risk (Anderson et al., 2003); see also Evans (2011), who links statistically significant intraday jumps to U.S. macro announcements.

³Volatility shocks are informative in predicting shifts in liquidity (Chordia et al., 2005).

cryptocurrency markets, which present unique challenges in measuring liquidity due to the fragmented nature of trading across multiple exchanges. Cryptocurrencies are traded on hundreds of platforms, each with its order book and liquidity profile. Furthermore, cryptocurrencies can be traded 24 hours a day, seven days a week against fiat currencies and other cryptocurrencies (Brauneis et al., 2021). This fragmentation requires empirical verification of models originally proposed for equity and traditional currencies markets to assess their applicability and accuracy in the cryptocurrency context.

Early work documented substantial transaction costs and wide effective spreads in Bitcoin, with costs varying systematically with market conditions and trading venues (Dyhrberg et al., 2018)⁴. Using exchange-level order-book data for Bitcoin to US dollar (BTCUSD), Brauneis et al. (2022b) show that liquidity is largely explained by exchange’s past liquidity, past market-wide liquidity and volatility, and fees charged on the blockchain for Bitcoin transfers. Consistent with this risk-based channel, robustness exercises that cast a wide net for candidate drivers typically find realized volatility to be the dominant determinant of illiquidity in Bitcoin (Ahmed, 2024)⁵. Fee-policy experiments further corroborate the microstructure mechanism: both maker–taker and zero-fee changes alter quoted and effective spreads, book depth, and execution quality in predictable ways, underscoring the role of market-making incentives in shaping tightness (Brauneis et al., 2022a; Galati, 2024).

Intraday studies document pronounced time-of-day and day-of-week periodicity in volatility and trading activity, concentrated when volatility and participation peak (Eross et al., 2019; Hansen et al., 2024a; Wang et al., 2020). A comprehensive cross-exchange analysis shows the global market “trades at tea time,” i.e., liquidity is poorest and volatility highest around overlapping European–U.S. hours—precisely when informed and institutional activity is likely most intense (Brauneis et al., 2025). Empirical studies show that liquidity

⁴(Dyhrberg et al., 2018) find that the highest trading activity, highest volatility and lowest spreads coincide with US market trading hours. The study examined Bitcoin on three exchanges— Gdax, Gemini, and Kraken —from September 7, 2017, to May 10, 2018, for fiat currency pairs.

⁵In the empirical analysis, (Ahmed, 2024) adopts the bid/ask spread estimator developed by Corwin and Schultz (2012) for 60-min high-frequency data on the low, high, opening, and closing exchange rates of Bitcoin against the US dollar.

costs in the US decline remarkably from the opening to the close of trading, creating an L-shaped curve (Amaya et al., 2018; Cattivelli and Pirino, 2019). Calendar studies also report marked seasonality in returns and volumes, implying non-uniform information arrival across the 24/7 clock and setting the stage for spread periodicity (Baur et al., 2019). At the same time, crypto’s fragmented venue structure and limits to cross-market arbitrage generate exchange-specific liquidity conditions and incomplete integration, so that spreads and depth differ across platforms and quote currencies even for the same underlying asset (Makarov and Schoar, 2020).

Finally, “macro” liquidity surveys indicate that more liquid cryptocurrencies exhibit greater efficiency and lower predictability, linking liquidity improvements to lower trading costs over time (Wei, 2018), while venue-selection and flow-of-funds within the network remain important in explaining cross-sectional differences in liquidity provision (Scharnowski, 2021). Zhang and Li (2023) show a negative relationship between liquidity and expected returns in the cryptocurrency market. Specifically, cryptocurrencies with higher liquidity in a given week tend to have lower returns in the following week. Han (2023) argues that cryptocurrencies with high liquidity risk (beta) earned a risk-adjusted return that was 4.4% higher per week than those with low liquidity risk, after controlling for market, size, and reversal factors. Zaremba et al. (2021) show a daily reversal effect, but the pattern is cross-sectional by liquidity, and the handful of the largest and most tradable coins show daily momentum rather than reversal. If alphas are concentrated in hard-to-trade assets and critically dependent on harvesting extreme returns on small, illiquid, and volatile coins⁶ (see also Cakici et al. (2024), an important consideration is whether these returns remain attractive after accounting for the higher transaction costs prevalent in this market. Additionally, the literature includes papers on applying machine learning methods to cryptocurrency price forecasting (Bouteska et al., 2024; Cheng et al., 2024; Maciel et al., 2022; Nasirtafreshi, 2022), while some works employ traditional financial econometrics models for liquidity forecasting

⁶More broadly, equity evidence shows that liquidity carries cross-sectional premia internationally, underscoring the economic relevance of precise liquidity measurement (Cakici and Zaremba, 2021).

(Fiszeder et al., 2024; Tzeng and Su, 2024).

Despite rapid progress, the crypto microstructure literature still leaves several questions open. Most studies either rely on best-quote or low-frequency spread proxies or examine volatility and volume periodicities without directly modeling periodicity in spreads across venues and quote currencies (Baur et al., 2019; Eross et al., 2019; Hansen et al., 2024a). While volatility emerges as a dominant determinant of illiquidity in Bitcoin and exchange design/fee policies measurably affect tightness and depth (Ahmed, 2024; Brauneis et al., 2022a,b; Galati, 2024), there is limited evidence on whether intraday periodic components in spreads remain incrementally informative once return dynamics and volatility are controlled, whether such periodicity materially weakens on weekends in a 24/7 market, and how strongly these features vary across major venues and quote currencies in a depth-adjusted setting. Fragmentation and incomplete cross-market integration imply that identical assets can exhibit exchange-specific tightness and depth (Makarov and Schoar, 2020), yet systematic, fixed-notional, cross-venue comparisons of high-frequency spreads remain scarce.

3 Data and Preliminary Results

3.1 Data Sources

The dataset used in this study consists of high-frequency order book data collected from two major cryptocurrency exchanges, Binance and Coinbase⁷. Following Hansen et al. (2024b), we focused on two major cryptocurrencies: Bitcoin (BTC) and Ethereum (ETH), selected for their dominant market positions and fundamental importance to the broader crypto ecosystem. Bitcoin remains the benchmark cryptocurrency with the largest market capitalization, while Ethereum is the primary platform for decentralized applications and smart contracts.

On Binance, these cryptocurrencies have been paired with three stablecoins that provide

⁷Empirically, liquidity varies across LOB levels—tightness and depth are distinct dimensions (Rakowski and Wang Beardsley, 2008)

significant liquidity in the ecosystem: Tether (USDT), USD Coin (USDC), and First Digital USD (FDUSD), resulting in six cryptocurrency-stablecoin trading pairs. In parallel, on Coinbase, we collected data for the same cryptocurrencies paired with USD, EUR, and USDT, providing an additional six trading pairs and enabling cross-exchange comparisons. For each trading pair, we collected the best bid and ask prices as well as order book depth information at 15-minute intervals, allowing us to calculate the mid-price and spread metrics. The data collection period spanned from July 1, 2024 until March 31, 2025.

A key aspect of our research design is the deliberate inclusion of different types of trading pairs across multiple exchanges. By examining both centralized exchange environments (Binance and Coinbase) and different quote currencies (USDT, USDC, FDUSD, USD, EUR), we can identify similarities and differences in market microstructure across these distinct liquidity environments. This comprehensive approach enables us to analyze liquidity patterns specifically within the cryptocurrency ecosystem, comparing how the same assets perform across different trading venues and against different stablecoins or fiat currencies.

3.2 Spreads Calculation

To ensure a robust measurement of liquidity in cryptocurrency markets, we employ a volume-weighted approach for calculating bid-ask spreads (Bianchi and Babiak, 2022). For our liquidity proxy, we use the Percent Quoted Spread (PQS) (Brauneis et al., 2022b), defined as:

$$\text{PQS}_t = \frac{P_{\text{ask},\text{VW}}^t - P_{\text{bid},\text{VW}}^t}{P_{\text{mid},\text{VW}}^t} \quad (1)$$

where $P_{\text{ask},\text{VW}}^t$ and $P_{\text{bid},\text{VW}}^t$ represent the volume-weighted ask and bid prices at time t , and $P_{\text{mid},\text{VW}}^t$ is the mid-price calculated as the average of volume-weighted bid and ask prices. Embedding depth via fixed notional book aggregation aligns the measure with multidimensional liquidity frameworks (Black et al., 2016). This approach is also similar to the method

used in Röscher and Kaserer (2013) and Griffith and Roseman (2019), where an order-size dependent, volume-weighted spread derived from the limit order book was used.

Unlike traditional spread calculations that rely solely on the best bid and ask prices, followed by Brauneis et al. (2022b), our methodology incorporates volume-weighted prices calculated based on all orders where the cumulative transaction volume exceeds USD 100,000⁸. Specifically:

$$P_{\text{bid},\text{VW}}^t = \frac{\sum_{i=1}^N P_{\text{bid},i}^t \times V_{\text{bid},i}^t}{\sum_{i=1}^N V_{\text{bid},i}^t} \quad (2)$$

$$P_{\text{ask},\text{VW}}^t = \frac{\sum_{i=1}^N P_{\text{ask},i}^t \times V_{\text{ask},i}^t}{\sum_{i=1}^N V_{\text{ask},i}^t} \quad (3)$$

where:

- $P_{\text{bid},i}^t$ and $P_{\text{ask},i}^t$ are the bid and ask prices at level i of the order book at time t .
- $V_{\text{bid},i}^t$ and $V_{\text{ask},i}^t$ are the corresponding bid and ask volumes at level i .

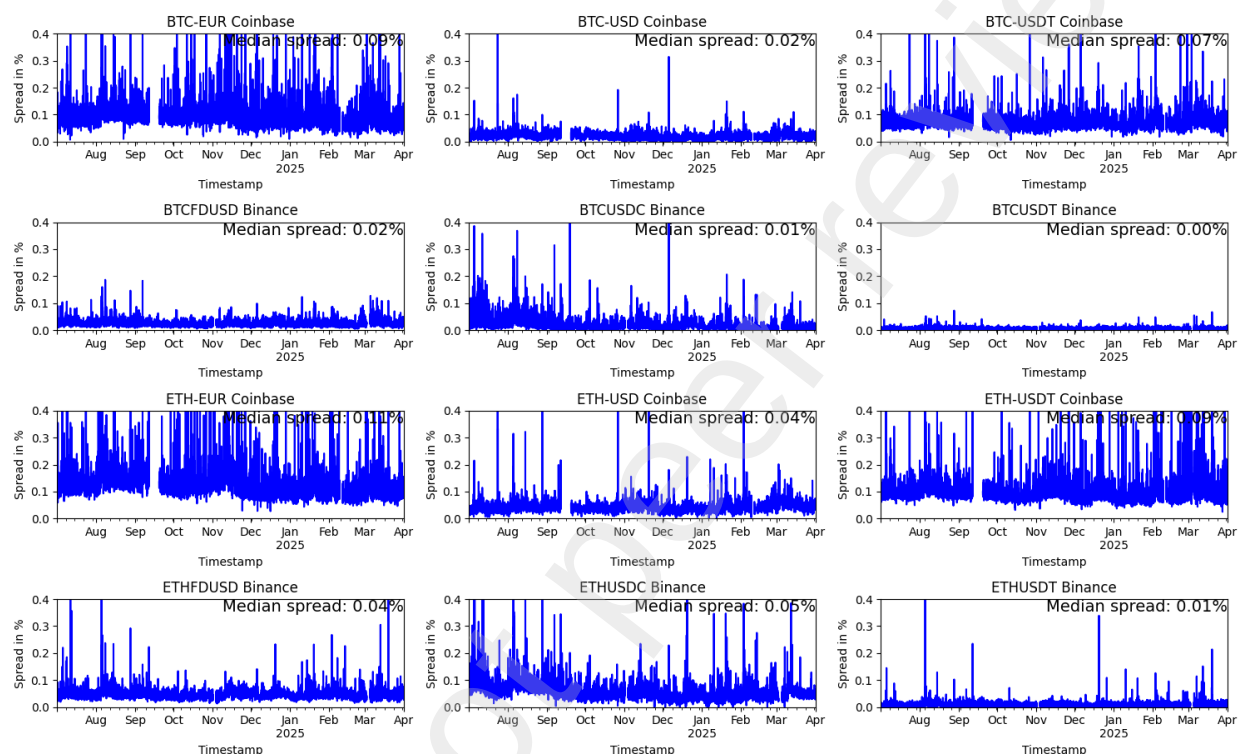
This volume-weighted approach offers several advantages for studying cryptocurrency markets. It provides a more accurate estimate of liquidity by filtering out small nominal trades and reducing bias from small trades that may not reflect true market conditions. It also facilitates comparability across cryptocurrency pairs and exchanges, ensuring consistency regardless of the specific asset being traded. We applied this methodology consistently across all cryptocurrency trading pairs from both Binance and Coinbase, collecting data at 15-minute intervals to capture intraday liquidity dynamics.

Figure 1 shows the time series of spreads for all 12 cryptocurrency markets in our study. Each chart displays the spread dynamics over time for a particular market, with the av-

⁸Our spread proxy aggregates quotes across the book until a fixed notional (USD 100k) is covered and computes a volume-weighted mid and spread. This ‘cost-at-size’ approach follows the same logic as the Xetra Liquidity Measure (XLM), which publishes round-trip costs for specified sizes (e.g., EUR100k) using multiple levels of the order book (https://xetra.com/xetra-de/instrumente/etfs-etps/statistics/xetra-liquiditytaetsmass-etfs?utm_source=chatgpt.com), which defines liquidity as the cost of an immediate institutional-size round-trip.

erage spread level shown in the upper right-hand corner. The figure highlights significant differences in both spread behavior and liquidity levels across cryptocurrency markets and exchanges.

Figure 1: Time Series of Spreads for Cryptocurrency Pairs



Note: The figure shows the time series of spreads for cryptocurrency pairs from both Binance (BTC and ETH paired with USDT, USDC, and FDUSD) and Coinbase (BTC and ETH paired with USD, EUR, and USDT). Data was collected at 15-minute intervals through March 31, 2025. Each chart shows the spread dynamics over time for a particular market, with the average spread level shown in the upper right-hand corner. The figure highlights significant differences in both spread behavior and liquidity levels across different trading pairs and exchanges.

3.3 Descriptive Statistics

Table 1 presents the descriptive statistics for the spread series across all 12 market pairs analyzed: twelve cryptocurrency pairs (from both Binance and Coinbase). The statistics include the number of observations (count), mean, standard deviation (sd), percentiles (1%, 25%, median, 75%, 99%), skewness, and kurtosis, with all values expressed as percentages (i.e., a value of 0.01 corresponds to 0.01%).

Table 1: Descriptive Statistics of Market Spreads for Cryptocurrency and Traditional Currency Pairs

Symbol	Count	Mean	Std	1%	25%	Median	75%	99%	Skew	Kurtosis
Panel A: Binance										
BTCFDUSD	25665	0.02	0.01	0.01	0.02	0.02	0.03	0.05	2.44	19.09
BTCUSDC	25665	0.02	0.02	0.00	0.01	0.01	0.02	0.09	5.05	60.98
BTCUSDT	25665	0.00	0.00	0.00	0.00	0.00	0.00	0.02	2.56	16.45
ETHFDUSD	25665	0.05	0.01	0.02	0.04	0.04	0.05	0.09	6.91	144.51
ETHUSDC	25665	0.05	0.03	0.02	0.04	0.05	0.06	0.14	4.99	75.83
ETHUSDT	25665	0.01	0.01	0.00	0.00	0.01	0.01	0.03	12.84	502.90
Panel A: Coinbase										
BTC-EUR	24835	0.09	0.06	0.04	0.07	0.09	0.10	0.23	21.54	755.67
BTC-USD	24835	0.02	0.01	0.00	0.02	0.02	0.03	0.05	10.46	459.05
BTC-USDT	24835	0.07	0.04	0.04	0.06	0.07	0.08	0.15	31.35	1414.19
ETH-EUR	24835	0.12	0.11	0.07	0.09	0.11	0.13	0.30	37.81	1876.99
ETH-USD	24835	0.04	0.02	0.02	0.03	0.04	0.05	0.08	29.50	1403.03
ETH-USDT	24835	0.09	0.07	0.05	0.08	0.09	0.10	0.21	55.68	5019.49

Note: This table presents descriptive statistics for percent quoted spreads (PQS) for cryptocurrency and traditional currency markets. The data spans through March 31, 2025, collected at 15-minute intervals. Panel A shows statistics for Bitcoin (BTC) and Ethereum (ETH) pairs on Binance with three stablecoins: Tether (USDT), USD Coin (USDC), and First Digital USD (FDUSD). Panel B shows the same cryptocurrencies on Coinbase paired with USD, EUR, and USDT.

Panel A shows that cryptocurrency markets on Binance generally exhibit extremely tight spreads, with means ranging from nearly 0.00% (BTCUSDT, BTCUSDC) to 0.01% (BTCFDUSD, ETHUSDC, ETHFDUSD). The data shows significant differences between trading pairs denominated in different stablecoins, with FDUSD pairs consistently showing wider spreads than their USDT counterparts. All Binance cryptocurrency pairs exhibit extremely high positive skewness (ranging from 4.95 to 35.98) and excess kurtosis (95.85 to 1881.17), indicating that their distributions are highly skewed to the right with frequent extreme values. Panel B reveals that cryptocurrency pairs on Coinbase generally have wider spreads than their Binance counterparts, with means ranging from 0.01% (BTC-USD, ETH-USD) to 0.04% (ETH-EUR, ETH-USDT). Notably, EUR-denominated pairs show consistently wider spreads than USD-denominated pairs on the same exchange. Similar to Binance, Coinbase pairs also exhibit extremely high skewness (2.63 to 48.12) and kurtosis (25.82 to 3191.46), with ETH-EUR showing the most extreme distributional characteristics. The observed data includes approximately 25,665 observations for Binance cryptocurrency pairs, 24,835 for Coinbase cryptocurrency pairs, and between 20,526 and 23,139 observations for traditional

currency pairs.

We perform several stationarity tests on our spread series, as reported in Table A.3 in Appendix A. The Augmented Dickey-Fuller (ADF), DF-GLS, Phillips-Perron and Zivot-Andrews tests consistently reject the null hypothesis of a unit root for all currency pairs, indicating the absence of stochastic trends (The Zivot-Andrews test, which allows for a possible structural break, also rejects the null hypothesis of a unit root in all cases). However, the KPSS test, which uses stationarity as the null hypothesis, shows mixed results, with some series potentially showing non-stationarity. As shown in Table A.4, both the Box-Pierce and Ljung-Box tests strongly reject the null hypothesis of no autocorrelation for all series, with particularly high test statistics for traditional currency pairs. These results confirm the presence of significant temporal dependencies in spread dynamics in both cryptocurrency and traditional currency markets, justifying our focus on periodicity patterns.

3.4 How Liquidity is Distributed over Time

We operationalize the above implications from the literature in three ways. First, we test the risk-pricing channel by estimating the relation between log-spreads and volatility proxies—squared returns and an EWMA volatility—expecting positive coefficients on both σ_t and r_t^2 . Second, we identify intraday periodicity in spreads via spectral methods and include the dominant harmonic as an explanatory factor; if spreads embody time-varying information and inventory premia, the periodic component should be statistically significant even after controlling for returns and volatility. Third, we exploit the 24/7 nature of crypto to test a clean weekday–weekend contrast: with lower information intensity on weekends, the weekend indicator should be associated with tighter spreads (*ceteris paribus*) and a reduced amplitude of the periodic component. Finally, consistent with fragmentation and market-design heterogeneity (Brauneis et al., 2022b; Makarov and Schoar, 2020), we examine cross-venue and quote-currency differences under a fixed notional; theory suggests that deeper venues and more liquid quotes (e.g., USD vs. EUR; USDT vs. FDUSD) should display systematically

tighter spreads, though the volatility–spread slope should remain positive across markets.

3.4.1 Intraday Patterns of Liquidity and Volatility

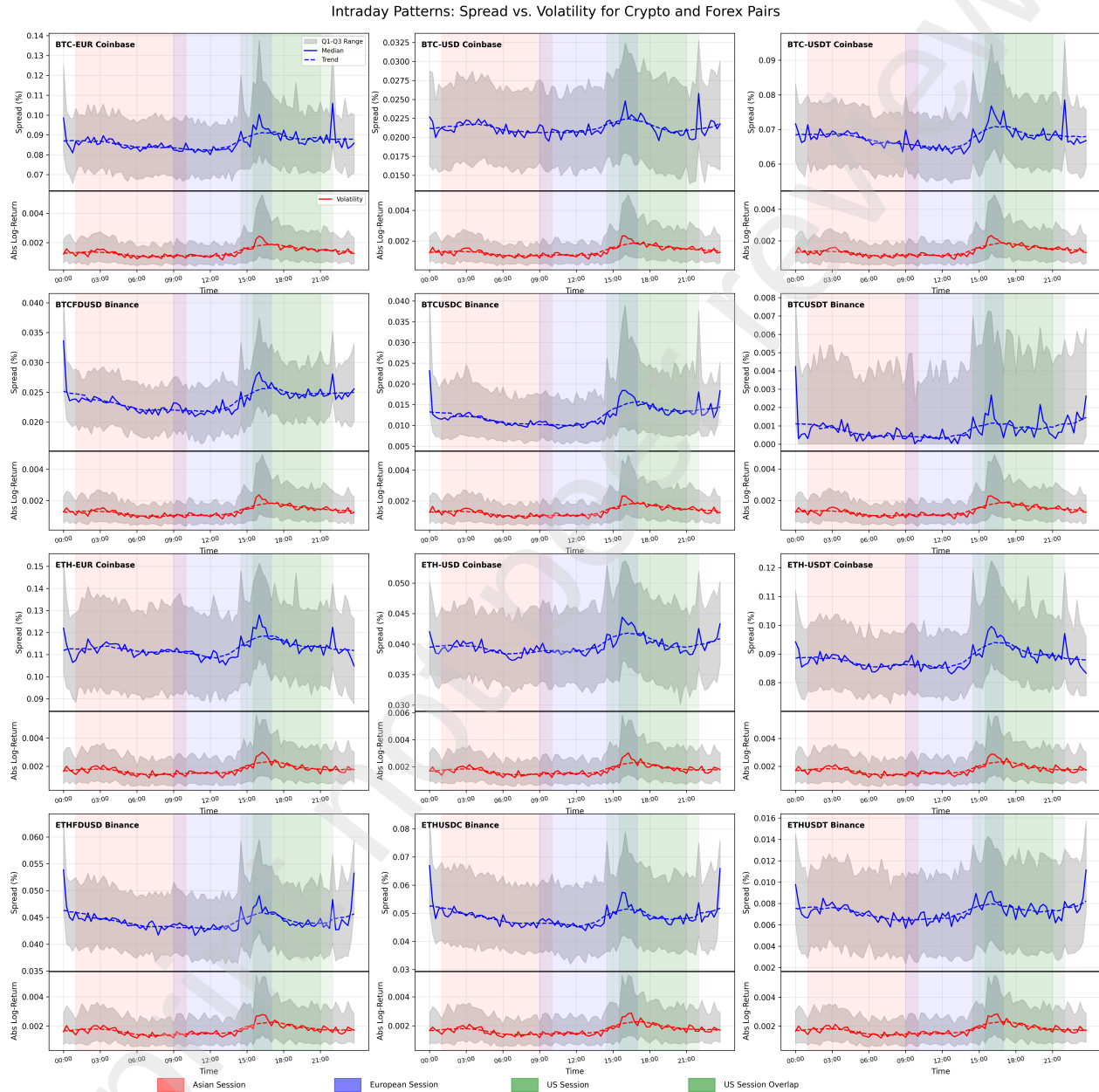
As the first step, we explore the relationship between market liquidity and volatility throughout the trading day. For each cryptocurrency pair, we organize observations into 15-minute time intervals across the trading day. For each time interval and trading pair, we compute three key statistical measures: the 25th percentile (Q1), median, and 75th percentile (Q3) of both spread and absolute log return values. This approach captures both the typical liquidity and volatility levels (median) and their dispersion (interquartile range) at different times of day, while minimizing the impact of outliers that frequently occur in high-frequency cryptocurrency data.

To extract the general trend that may be obscured by time-specific noise, we apply a centered moving average to the median values. For each time of day, we calculate a 15-period window (corresponding to a 3.75-hour interval) that smooths out short-term fluctuations while preserving the underlying periodic patterns.

Figure 2 presents a paired visualization where each cryptocurrency pair’s spread (top panel) and volatility (bottom panel) are displayed directly above one another. The gray shaded areas indicate the interquartile range (25th to 75th percentile) for both metrics, capturing the typical variation around the median. To facilitate interpretation, we overlay time zone bands corresponding to primary global market hours: Asian markets (01:00-10:00 CET, light red), European markets (09:00-17:00 CET, light blue), and US markets (14:30-22:00 CET, light green).

Our analysis of intraday liquidity and volatility patterns reveals important insights into cryptocurrency market microstructure. For cryptocurrency pairs on both Coinbase and Binance, we observe a distinct inverted sinusoidal pattern in spreads, characterized by the lowest spreads during Asian and early European market hours, followed by a rapid transition that peaks at the beginning of the US trading session. Volatility follows a remarkably similar

Figure 2: Joint Intraday Analysis of Spreads and Volatility for Cryptocurrency Pairs



Note: For each cryptocurrency pair, the top panel shows the median spread (blue) and the bottom panel shows the median volatility (red). Both panels include the interquartile range (gray shaded area) and a 15-period moving average (dashed lines). Background colors indicate primary market hours in CET: Asian markets (01:00-10:00, light red), European markets (09:00-17:00, light blue), and US markets (14:30-22:00, light green). The analysis covers all 12 cryptocurrency pairs across Binance and Coinbase exchanges.

pattern, with price fluctuations rising exactly as spreads increase.

This strong temporal correspondence suggests that liquidity in cryptocurrency markets responds directly to changes in market volatility, regardless of whether the quoted currency is a fiat currency (USD, EUR) or a stablecoin (USDT, USDC, FDUSD). The synchronized movement between spreads and volatility across nearly all cryptocurrency markets supports the hypothesis of market-making risk models, where liquidity providers require wider spreads as compensation for taking positions during periods of higher price uncertainty.

This relationship appears to be particularly strong in cryptocurrency markets, suggesting that despite their 24/7 operation, these markets experience significant intraday fluctuations in liquidity that are closely tied to global trading patterns and volatility regimes. Interestingly, we observe consistent patterns across both major cryptocurrencies (BTC and ETH) and across different exchanges (Binance and Coinbase), indicating that these liquidity dynamics may be fundamental characteristics of cryptocurrency markets rather than exchange-specific phenomena.

Overall, the intraday profiles we document are consistent with microstructure predictions that inventory and information premia rise when volatility and information arrival intensify, yielding wider spreads even amid active quote competition (Copeland and Galai, 1983; Hasbrouck, 1991; Kyle, 1985). The peak into U.S. hours accords with evidence on time-of-day periodicities in crypto activity and risk (Baur et al., 2019; Eross et al., 2019; Hansen et al., 2024a) and with cross-exchange results that liquidity is poorest and volatility highest around the Europe–U.S. overlap (Brauneis et al., 2025).

3.4.2 Weekend vs. Weekday Patterns in Cryptocurrency Markets

One of the distinct features of cryptocurrency markets is their continuous 24/7 operation, which allows us to examine the contrast between weekend and weekday trading conditions. Figure 3 presents a detailed comparison of intraday spread and volatility patterns between weekdays (blue) and weekends (green) for twelve cryptocurrency pairs across Binance and

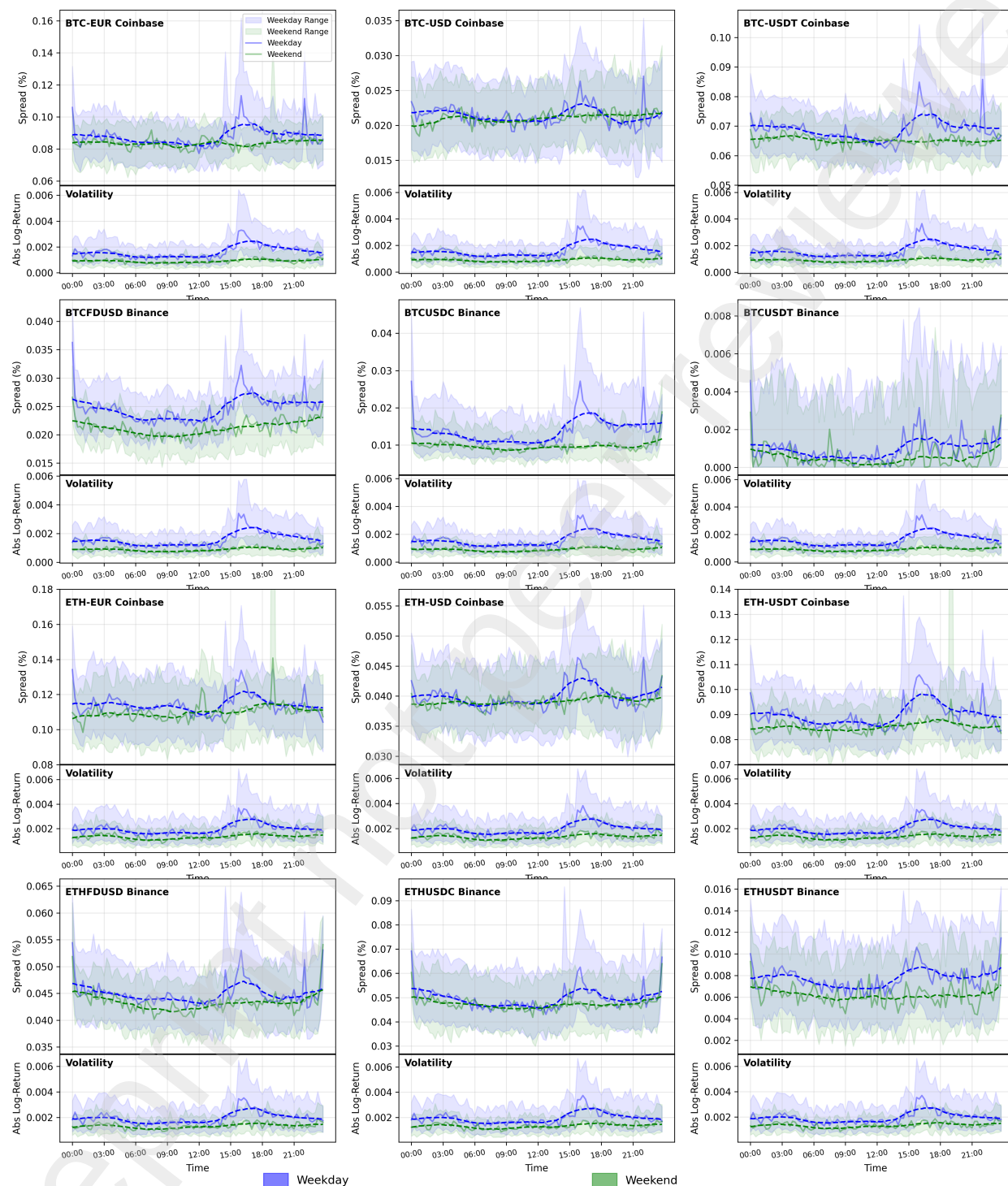
Coinbase exchanges.

Our analysis reveals a notable contrast between weekday and weekend trading dynamics in cryptocurrency markets, consistent with prior evidence on day-of-week and time-of-day regularities in crypto activity, volatility and liquidity (Baur et al., 2019; Brauneis et al., 2025; Eross et al., 2019; Hansen et al., 2024a). First, spreads are consistently lower on weekends for nearly all cryptocurrency pairs, with the gap being most pronounced during periods that coincide with typical global trading hours (15:00–18:00 CET). This pattern is particularly evident for the BTC-EUR, BTC-USD and BTCFDUSD pairs, where median spreads can be 20–50% higher on weekdays than on weekends during peak hours. Importantly, the weekend spread narrowing is accompanied by a corresponding decrease in volatility. For all cryptocurrency pairs, weekend median volatility is lower than weekday volatility, especially during these hours, in line with documented calendar effects in crypto (Baur et al., 2019) and the alignment of crypto’s intraday risk with European–U.S. trading windows (Hansen et al., 2024a; Wang et al., 2020). This parallel movement reinforces our earlier finding that liquidity provision in cryptocurrency markets is closely linked to volatility, as predicted by inventory- and information-risk models of the spread (Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997).

Second, weekend spreads display a substantially flattened pattern with minimal intraday variation. The characteristic peaks observed during weekday trading hours are almost entirely absent on weekends, resulting in a relatively stable spread throughout the day. Weekend volatility is remarkably flat across all hours, appearing as an almost horizontal line for most pairs. This attenuation of intraday periodicity on weekends echoes prior evidence that crypto’s strongest time-of-day patterns coincide with European–U.S. sessions on weekdays and weaken outside those windows (Baur et al., 2019; Hansen et al., 2024a; Wang et al., 2020).

These findings indicate that despite the 24/7 nature of cryptocurrency markets, their microstructure dynamics remain heavily influenced by global trading patterns. The near-

Figure 3: Comparison of Weekday and Weekend Liquidity and Volatility Patterns for Cryptocurrencies



Note: For each cryptocurrency pair, the top panel shows spread patterns and the bottom panel shows volatility (absolute log returns). Blue lines and shading represent weekday patterns (Monday-Friday), while green represents weekend patterns (Saturday-Sunday). Solid lines show median values, dashed lines show 15-period moving averages, and shaded areas represent the interquartile range. The analysis covers six pairs each from Binance and Coinbase exchanges.

complete flattening of intraday patterns during weekends strongly suggests that the characteristic volatility and spread fluctuations observed during weekdays are largely driven by information-intensive flow concentrated in European–U.S. hours, consistent with cross-exchange evidence that liquidity is poorest and volatility highest around the Europe–U.S. overlap (Brauneis et al., 2025) and with network- and exchange-level analyses of time-of-day intensity (Jahanshahloo et al., 2022; Wang et al., 2020).

4 Relative analysis

4.1 Relative Volatility and Spread for Hour-of-the-Day

To examine the intraday behavior of market liquidity and volatility, we use a relative analysis technique that examines how these measures vary over different time periods. Unlike absolute measures, this approach normalizes values against recent history, allowing direct comparisons between different market pairs despite their different starting levels. Our first step was to examine how volatility and spread behave at different hours of the day relative to other hours within the same day. Following Hansen et al. (2024b), we compute hour-specific relative measures for both volatility and spread:

$$\lambda_{\sigma}^{\text{hour}}(h) = \frac{1}{n_h} \sum_{w,d} \frac{|y_{\tau(w,d,h)}|}{\sum_{j=0}^{23} |y_{\tau(w,d,h-j)}|} \quad (4)$$

$$\lambda_s^{\text{hour}}(h) = \frac{1}{n_h} \sum_{w,d} \frac{s_{\tau(w,d,h)}}{\sum_{j=0}^{23} s_{\tau(w,d,h-j)}} \quad (5)$$

where:

- h – hour of the day, $h = 0, \dots, 23$
- w – week number in our sample
- d – day of the week
- $y_{\tau(w,d,h)}$ – absolute return at time point (w, d, h)

- $s_{\tau(w,d,h)}$ – spread at time point (w, d, h)
- $n_h \approx 7 \times W$ – number of observations per hour, where W is the total number of weeks in our sample

The interpretation of these measures is straightforward - if $\lambda_{\sigma}^{\text{hour}}(h) > 1$, volatility is higher at hour h compared to the daily average. If $\lambda_s^{\text{hour}}(h) > 1$, the spread is higher at hour h , indicating lower liquidity. This hourly analysis allows us to identify specific times of day when liquidity conditions systematically deviate from the daily average, which is particularly important for understanding market microstructure in 24/7 cryptocurrency markets.

Figure 4 shows the results of this analysis across all market pairs in polar coordinates, where the angle represents the hour of the day and the distance from the center represents the relative measure. Cryptocurrency pairs show remarkably similar patterns across exchanges and pairs, with volatility and spreads forming clear directional patterns around US trading hours (15:00-18:00 CET).

The hour-normalized profiles confirm, in the spirit of [Hansen et al. \(2024b\)](#), that both volatility and spreads load most heavily into the U.S. session, with a common directional pattern across assets, venues, and quote currencies. The tight co-movement between $\lambda_{\sigma}^{\text{hour}}(h)$ and $\lambda_s^{\text{hour}}(h)$ is exactly what inventory- and information-risk models predict when information arrival and return variance rise, raising the cost of immediacy ([Glosten and Milgrom, 1985](#); [Ho and Stoll, 1981](#); [Huang and Stoll, 1997](#)).

4.2 Relative Volatility and Spread for Day-of-the-Week

Building on our hour-of-the-day analysis, we apply a similar normalization approach to examine day-of-the-week patterns of volatility and spreads, and see if both characteristics are lower during the weekend. For each day of the week, we compute a ratio measure that compares the sum of the spread (or volatility) on that day to the sum over the previous seven days. Formally, for each market pair, we calculate:

Figure 4: Polar Representation of Hour-of-Day Relative Measures



Note: Each chart shows the relative volatility measure (red bars) and spread measure (blue line) in polar coordinates. The radial axis represents the ratio value, with the circle at 1.0 indicating the average level. The angular coordinate represents the hour of the day, with midnight at the top and proceeding clockwise.

$$\lambda_{\sigma}^{\text{day}}(d) = \frac{1}{n_d} \sum_w \frac{|y_{\tau(w,d,h)}|}{\sum_{i=0}^6 |y_{\tau(w,d-i,h)}|} \quad (6)$$

$$\lambda_s^{\text{day}}(d) = \frac{1}{n_d} \sum_w \frac{s_{\tau(w,d,h)}}{\sum_{i=0}^6 s_{\tau(w,d-i,h)}} \quad (7)$$

where:

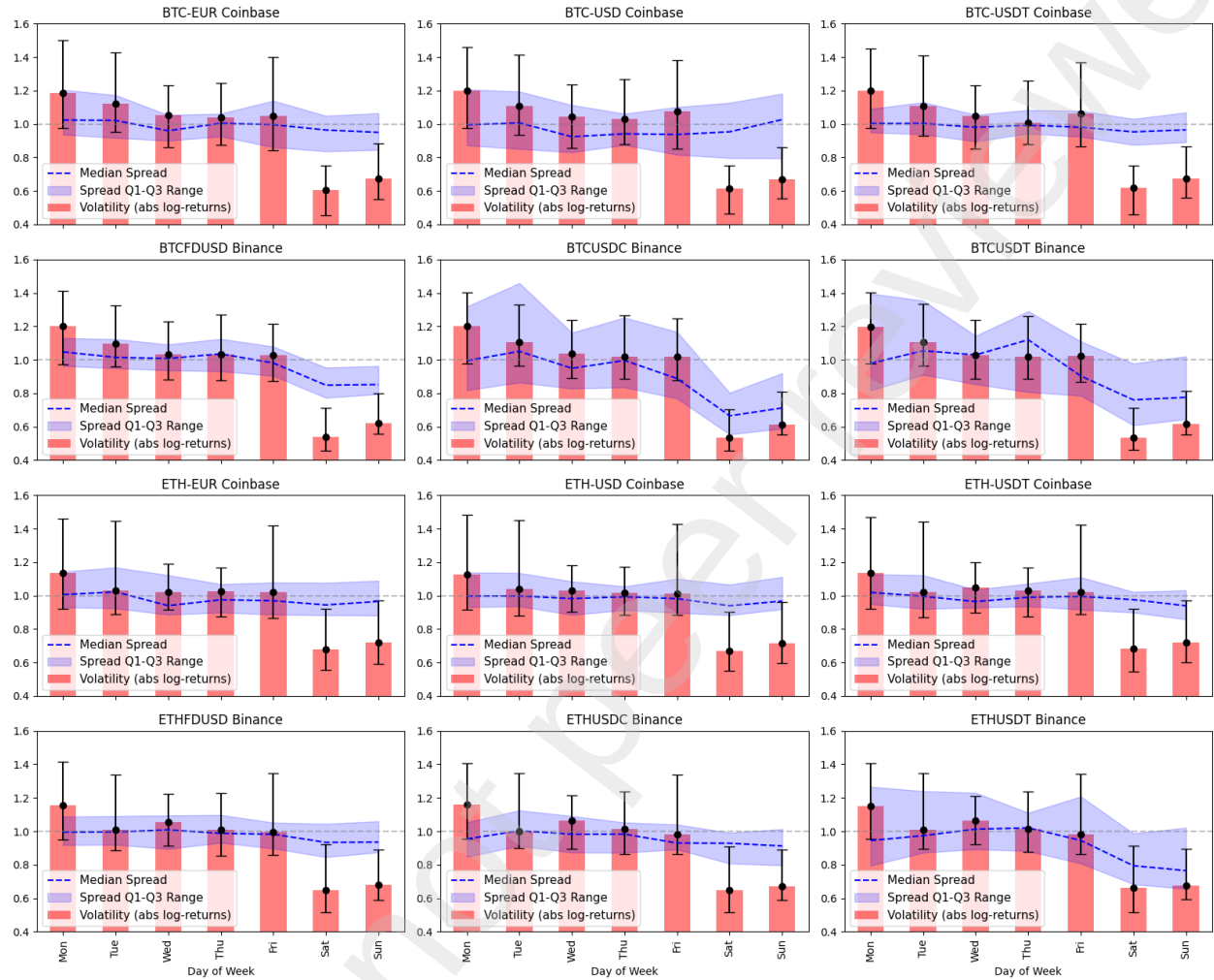
- d – day of the week, $d = 1, \dots, 7$
- w – week number
- h – hour of the day
- $y_{\tau(w,d,h)}$ – absolute return at time point (w, d, h)
- $s_{\tau(w,d,h)}$ – spread at time point (w, d, h)
- $n_d \approx W$ – number of observations per day of the week

If $\lambda_\sigma^{\text{day}}(d) > 1$, volatility is higher on day d compared to the weekly average. If $\lambda_s^{\text{day}}(d) > 1$, the spread is higher on day d , indicating lower liquidity. Figure 5 presents the results of this analysis for selected cryptocurrency pairs. For cryptocurrency pairs, weekdays (Monday through Friday) show relatively stable volatility and spread measures near the neutral threshold, while weekends show low volatility (often 30-40% below average).

The day-normalized measures show pronounced weekend compression: both $\lambda_\sigma^{\text{day}}(d)$ and $\lambda_s^{\text{day}}(d)$ fall below one on Saturday–Sunday, indicating lower volatility and tighter spreads and a marked attenuation of the intraday cycle. This weekday–weekend contrast is consistent with crypto calendar regularities and weaker time-of-day effects on weekends (Baur et al., 2019; Eross et al., 2019; Wang et al., 2020) and with the periodicity in volatility and liquidity centered on European–U.S. hours (Brauneis et al., 2025; Hansen et al., 2024b). The mechanism aligns with microstructure theory: reduced information intensity lowers adverse-selection and inventory premia, tightening spreads even if some participation persists (Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997). Relative normalization across markets is key here, revealing a common weekend flattening in spreads — not just returns or volumes—across venues and quote currencies.

A visual examination reveals that weekend effects exist, but they are modest and vary

Figure 5: Day-of-Week Relative Volatility and Spread Measures



Note: The bars (red) represent the relative volatility measure for each day of the week, with error bars showing the statistical uncertainty (Q1 to Q3 range). The blue dashed line shows the corresponding relative spread measure. The horizontal line at 1.0 indicates the average level. Values above 1.0 indicate higher-than-average volatility or spreads for that day.

across assets, venues, and quote currencies. Typically, the median spread and volatility decrease on Saturday and Sunday, though the interquartile ranges often overlap with weekday values. This weaker signal aligns with previous findings that day-of-week patterns in cryptocurrency are less stable and smaller in magnitude than the significant time-of-day cycle (Baur et al., 2019; Eross et al., 2019; Wang et al., 2020). It also accords with Hansen et al. (2024b), who document that the dominant periodicity is daily (aligned with European–U.S.

hours) rather than strictly weekly. Heterogeneity across exchanges and quote currencies is in line with fragmentation and venue-specific microstructure documented for crypto markets (Brauneis et al., 2022b; Makarov and Schoar, 2020). Taken together, the day-of-week normalization suggests a general but not universal weekend attenuation of spreads and volatility.

5 Fast Fourier Transform Analysis and Harmonic Modeling

To rigorously characterize the underlying periodicities in the spread time series, apply spectral analysis techniques to logarithmic bid-ask spreads. We employ a three-stage approach: (1) objectively identify the dominant frequencies in liquidity patterns, (2) estimate the phase relationships that determine cycle timing, and (3) statistically validate the significance of these cyclical components using the complete time series data.

We begin by computing the median log-spread values for each time point across the day, resulting in a 96-element vector (representing 15-minute intervals). Fast Fourier Transform (FFT) is then applied to this median pattern to decompose it into its frequency components. FFT efficiently transforms time-domain data into the frequency domain by expressing the signal as a sum of sinusoidal components of varying frequencies and amplitudes:

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-i2\pi kn/N} \quad (8)$$

where $X(k)$ represents the spectral components, $x(n)$ is our time-series data, and N is the number of time points. By analyzing the magnitude of these spectral components, we identify the dominant frequencies $f_1, f_2, \dots, f_n$ ranked by their power in the spectrum. These frequencies represent the most important cyclical patterns in the intraday spread data.

Using the dominant frequencies identified in Step 1, we fit two harmonic regression models to the median log-spread pattern:

1. Single-frequency model:

$$\log(S_t) = a_0 + A_1 \sin(2\pi f_1 t + \phi_1) + \varepsilon_t \quad (9)$$

2. Two-frequency model:

$$\log(S_t) = a_0 + A_1 \sin(2\pi f_1 t + \phi_1) + A_2 \sin(2\pi f_2 t + \phi_2) + \varepsilon_t \quad (10)$$

where a_0 is the intercept, A_1 and A_2 are amplitude parameters, and ϕ_1 and ϕ_2 are phase shift parameters that determine the timing of peaks and troughs in the cyclical pattern⁹. The phase parameters ϕ_1 and ϕ_2 are particularly important as they capture the timing of liquidity cycles throughout the day. These phase shifts allow us to determine when spreads tend to widen or narrow within the identified periodic cycles.

Finally, to validate the periodicity patterns identified in the previous step and ensure they represent genuine cyclical behavior rather than random fluctuations, we extend our analysis beyond the median pattern to the full log-spread time series. We implement a formal statistical testing framework using harmonic regression. We first convert timestamps into normalized time to standardize the analysis across different market pairs:

$$t_{\text{norm}} = \frac{\text{hours} + \text{minutes}/60}{24} \quad (11)$$

Next, using the frequencies and estimated phase parameters from Steps 1 and 2, we construct explanatory variables of the form:

$$X_1(t) = \sin(2\pi f_1 t + \hat{\phi}_1) \quad (12)$$

$$X_2(t) = \sin(2\pi f_2 t + \hat{\phi}_2) \quad (13)$$

⁹We estimate these parameters using non-linear least squares optimization with the the function, which minimizes the sum of squared residuals between the observed median pattern and the harmonic model

where $\hat{\phi}_1$ and $\hat{\phi}_2$ are the phase parameters estimated in Step 2. We then perform linear regression on the complete time series of log-spreads:

$$\log(S_t) = \beta_0 + \beta_1 X_1(t) + \beta_2 X_2(t) + \epsilon_t \quad (14)$$

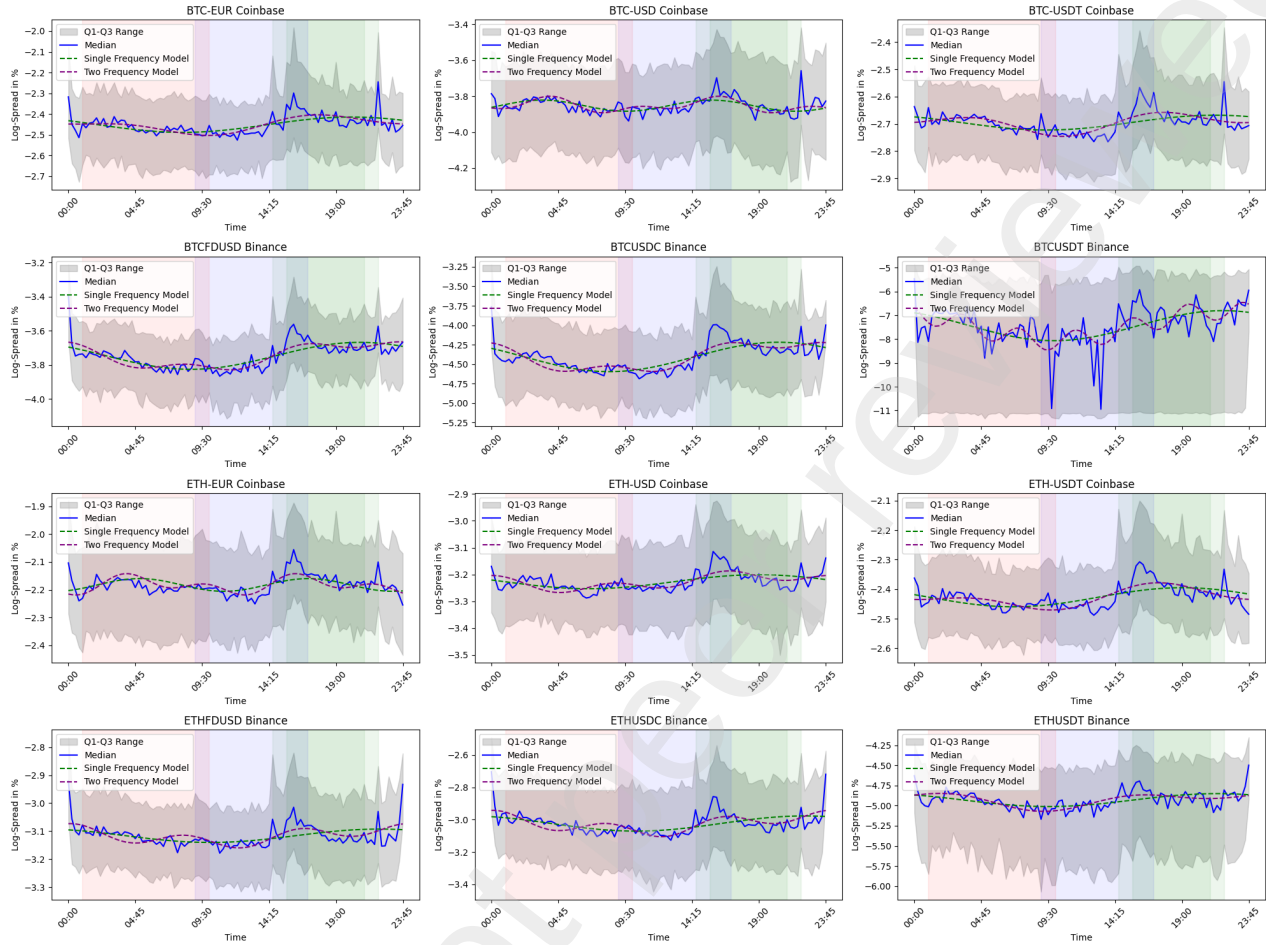
This allows us to apply standard statistical inference techniques to assess the significance of the periodic components. This regression uses heteroskedasticity and autocorrelation consistent (HAC) standard errors to account for potential serial correlation in the residuals. The t-statistics and p-values from this regression provide formal statistical tests of the significance of the identified cyclical components across the entire sample period, not just in the median pattern. Additionally, we evaluate the overall explanatory power of the periodic components using the coefficient of determination (R^2).

To visualize the fit of our harmonic models, Figure 6 presents the results of both single-frequency and two-frequency harmonic regressions for each market pair in our sample. The figure displays the median log-spread across the trading day (blue line) along with the interquartile range (gray shaded area), and overlays the fitted values from the single-frequency model (green dashed line) and the two-frequency model (purple dashed line).

Table 2 presents the results of our harmonic regression analysis. The table reports the dominant frequencies, phase shifts, parameter estimates, and corresponding t-statistics for both single-frequency and two-frequency models across all cryptocurrency assets in our sample. A notable pattern emerges: almost all cryptocurrency assets exhibit a primary cyclical component with frequency of approximately 0.01, corresponding to a daily (24-hour) cycle. This consistency suggests a strong global trading pattern affects cryptocurrency liquidity regardless of trading venue or specific asset. The t-statistics reported in Table ?? reveal that the cyclical components are highly significant for all cryptocurrency assets.

Our analysis reveals several insights about cryptocurrency market microstructure that are consistent with, and extend, the existing literature. First, the majority of pairs exhibit a

Figure 6: Harmonic Model Fit to Intraday Log-Spreads.



Note: This figure shows the median log-spread pattern across the trading day (blue solid line) with the interquartile range (gray shaded area). The green dashed line represents the fitted values from the single-frequency harmonic model, while the purple dashed line shows the fitted values from the two-frequency model. Colored bands highlight major trading sessions: Asian (red), European (blue), and US (green).

dominant cycle with frequency about 0.01 (one cycle per day), indicating that—despite 24/7 trading—liquidity follows a strong daily periodicity. This aligns with evidence that crypto volatility and liquidity display pronounced intraday seasonality centered on European–U.S. hours (Hansen et al., 2024b) and echoes classic intraday regularities documented in other markets (e.g., Mcinish and Wood, 1992). The phase estimates are remarkably similar across assets and venues, suggesting synchronized timing of liquidity provision across markets, consistent with commonality in liquidity (Hasbrouck, 2001) and strategic clustering of order flow at focal times (Admati and Pfleiderer, 1988).

Table 2: Harmonic Regression Results for Intraday Spread Periodicity

Symbol	Single-frequency harmonic model					Two-frequency harmonic model							N_{obs}
	f_1	ϕ	β_1	T_{β_1}	R^2	f_1	β_1	T_{β_1}	f_2	β_2	T_{β_2}	R^2	
Panel A: Binance													
BTCFDUSD	0.01	-0.71	-0.08	<u>-28.49</u>	0.03	0.01	-0.08	<u>-23.84</u>	0.03	0.03	<u>7.68</u>	0.03	25665
BTCUSDC	0.01	-0.62	-0.18	<u>-21.35</u>	0.02	0.01	-0.18	<u>-16.87</u>	0.03	0.08	<u>6.85</u>	0.02	25665
BTCUSDT	0.01	2.12	0.26	<u>10.73</u>	0.00	0.01	0.26	<u>10.16</u>	0.06	-0.17	<u>-6.53</u>	0.01	25665
ETHFDUSD	0.01	-1.07	-0.03	<u>-12.76</u>	0.01	0.01	-0.03	<u>-10.37</u>	0.03	0.03	<u>9.05</u>	0.01	25665
ETHUSDC	0.01	-1.12	-0.05	<u>-13.31</u>	0.01	0.01	-0.05	<u>-10.31</u>	0.03	0.04	<u>8.24</u>	0.01	25665
ETHUSDT	0.01	-0.92	-0.09	<u>-8.98</u>	0.00	0.01	-0.09	<u>-7.95</u>	0.02	0.06	<u>5.27</u>	0.00	25665
Panel B: Coinbase													
BTC-EUR	0.01	-0.55	-0.05	<u>-17.31</u>	0.01	0.01	-0.05	<u>-13.91</u>	0.02	0.02	<u>6.69</u>	0.01	24835
BTC-USD	0.02	-0.31	0.04	<u>8.80</u>	0.00	0.02	0.04	<u>7.51</u>	0.04	-0.03	<u>-4.31</u>	0.00	24835
BTC-USDT	0.01	-0.88	-0.04	<u>-16.53</u>	0.01	0.01	-0.04	<u>-13.35</u>	0.02	0.03	<u>9.64</u>	0.02	24835
ETH-EUR	0.02	-1.07	0.02	<u>8.83</u>	0.00	0.02	0.02	<u>6.83</u>	0.04	-0.02	<u>-4.29</u>	0.00	24835
ETH-USD	0.01	-0.25	-0.02	<u>-7.05</u>	0.00	0.01	-0.02	<u>-5.54</u>	0.03	0.02	<u>5.75</u>	0.00	24835
ETH-USDT	0.01	-0.27	-0.05	<u>-20.03</u>	0.02	0.01	-0.05	<u>-15.77</u>	0.02	0.03	<u>9.90</u>	0.02	24835

Note: The table presents harmonic regression results for log-transformed spreads using two model specifications. The first model includes a single-frequency component, while the second incorporates two frequency components. Bold and underlined values indicate t-statistics with absolute values greater than 3, representing statistical significance at approximately the 0.1% level. T values represent t-statistics for the amplitude parameters. Freq represents the identified dominant frequencies from Fourier analysis, measured in cycles per day. R^2 represents the coefficient of determination for each model. ϕ represents the estimated phase shift in radians.

Second, the amplitude parameters on the primary (and, for many pairs, secondary) harmonics are highly significant—often with $|t| > 15$ —which indicates that these periodic components are genuine features of the data rather than artefacts of averaging. This mirrors results in the intraday seasonality literature, where deterministic periodic factors are statistically powerful in high-frequency settings even when estimated in simple harmonic or flexible Fourier forms (Andersen and Bollerslev, 1997). In our context, the statistical strength of the periodic terms is precisely what one would expect if inventory- and information-risk premia vary systematically over the trading day (Glosten and Milgrom, 1985; Ho and Stoll, 1981; Huang and Stoll, 1997).

Third, secondary frequencies typically fall in the 0.02–0.03 range (roughly 8–12 hours), which plausibly reflects the influence of major session transitions and overlaps (Asia–Europe, Europe–U.S.) on liquidity provision. This interpretation is consistent with crypto time-of-day evidence that activity and risk concentrate around the Europe–U.S. window (Brauneis et al., 2025; Wang et al., 2020) and with broader findings on calendar and intraday seasonality in crypto (Baur et al., 2019; Eross et al., 2019).

Finally, while the cyclical terms are highly significant, the overall explanatory power of the parsimonious harmonic regressions is modest (R^2 of 0.00–0.03). This is in line with prior work showing that intraday seasonals, though crucial for identification and forecasting, explain only a limited share of unconditional variation in high-frequency measures (Andersen and Bollerslev, 1997). In our case, the result underscores that periodicity is one important component of spreads, alongside volatility and return dynamics; the consistent detection of significant cycles across Binance and Coinbase and across BTC and ETH with multiple quote currencies suggests that periodic liquidity fluctuations are a market-wide feature of crypto microstructure rather than an exchange-specific artefact (Hansen et al., 2024b; Hasbrouck, 2001).

5.1 Integrated Model

Having shown the existence of significant cyclical patterns in liquidity through our FFT analysis, we now incorporate these periodic components into a comprehensive model of spread determinants. We construct a regression framework that captures multiple dimensions of market conditions that may affect liquidity:

$$\log(S_t) = \alpha + \sum_{j=0}^3 \beta_j r_{t-j} + \gamma_1 r_t^2 + \gamma_2 \sigma_t + \delta \cdot FFT_t + \varepsilon_t \quad (15)$$

where $\log(S_t)$ is the logarithm of the bid-ask spread at time t , r_{t-j} represents returns at various lags (with $j = 0$ being contemporaneous returns), r_t^2 is the squared return as a measure of instantaneous volatility, σ_t is the exponentially weighted moving average (EWMA) volatility with a decay factor of 0.1, and FFT_t is the periodic component identified through our Fourier analysis, defined as $\sin(2\pi f_1 t + \hat{\phi}_1)$. This specification allows us to disentangle the effects of market returns, volatility dynamics, and cyclical patterns on market liquidity.

The model incorporates three key categories of explanatory variables:

1. **Return dynamics:** Current and lagged returns ($r_t, r_{t-1}, r_{t-2}, r_{t-3}$) capture the rela-

tionship between price movements and liquidity, including potential asymmetric responses to positive versus negative returns.

2. **Volatility measures:** Two complementary measures of volatility are included: squared returns (r_t^2) as a high-frequency indicator of instantaneous volatility, and EWMA volatility (σ_t) as a smoothed measure of sustained volatility.
3. **Periodicity component:** The FFT_t variable incorporates the dominant cyclical pattern identified through our Fourier analysis, enabling us to capture the systematic intraday variations in liquidity that follow regular temporal patterns.

To facilitate direct comparison of coefficient magnitudes across variables and market pairs, all explanatory variables are standardized to have a mean of zero and a standard deviation of one. This standardization ensures that the reported coefficients represent the effect of a one standard deviation change in the explanatory variable on the log spread.

To ensure robust statistical inference, we estimate this model using heteroskedasticity and autocorrelation consistent (HAC) standard errors with Newey-West correction. This approach accounts for potential serial correlation and time-varying volatility in the residuals. Additionally, we apply a Bonferroni correction to our significance threshold to address the multiple testing problem arising from analyzing 18 different market pairs simultaneously. Specifically, we divide the standard 2.5% significance level by 12, resulting in an adjusted significance level of approximately 0.00139. This corresponds to a t-statistic threshold of approximately 3. Consequently, in Table 3, we highlight coefficients with t-statistics exceeding 3 in absolute value as statistically significant.

Table 3 presents the regression results for all market pairs in our sample. Several key patterns emerge across different cryptocurrency pairs and exchanges. First, volatility emerges as the strongest determinant of spreads for all cryptocurrency pairs. The EWMA volatility measure (σ_t) shows particularly large and highly significant positive coefficients, ranging from 0.18 for BTCUSDT on Binance to 0.47 for ETH-USDT on Coinbase. This strong positive relationship indicates that sustained periods of high volatility significantly increase spreads,

consistent with increased risk for market makers during volatile conditions. The instantaneous volatility measure (r_t^2) shows more mixed effects, with significant positive coefficients for several Bitcoin pairs (particularly on Coinbase), but weaker effects for other pairs. This suggests that the persistent component of volatility (captured by the EWMA) is a more consistent determinant of liquidity than instantaneous volatility spikes.

Second, we observe consistently negative and statistically significant coefficients on both contemporaneous and lagged returns across most cryptocurrency pairs. This inverse relationship between returns and spreads suggests that liquidity tends to decrease (spreads widen) during periods of price declines. The effect is particularly pronounced for ETH pairs, where the contemporaneous return coefficient reaches -0.06 for ETH-USD on Coinbase (t-statistic: -6.38). Lagged returns also show significant negative relationships with spreads, especially in the first lag (r_{t-1}), indicating persistent effects of price movements on liquidity. This finding suggests that cryptocurrency market makers may adjust their liquidity provision asymmetrically in response to price movements, potentially reflecting risk management considerations during market downturns.

Third, the periodicity component (FFT) shows statistical significance for several cryptocurrency pairs, but with varying directions and magnitudes. For Binance, BTCFDUSD, BTCUSDC, and ETHUSDC show significant negative coefficients, while BTCUSDT shows a significant positive coefficient. On Coinbase, BTC-EUR shows a significant negative coefficient, while BTC-USD and ETH-EUR show significant positive coefficients. These mixed effects suggest that while cyclical patterns are present in cryptocurrency liquidity, their influence varies across different trading pairs and exchanges. The fact that some pairs show negative coefficients (indicating tighter spreads during peak cycle times) while others show positive coefficients highlights the complexity of intraday liquidity patterns in cryptocurrency markets.

Fourth, the weekend dummy variable reveals interesting patterns in weekend trading behavior. Six out of the twelve cryptocurrency pairs show statistically significant weekend

effects, but with differing signs. BTC-USD, BTC-USDT, and ETH-USD on Coinbase, along with ETHFDUSD on Binance, show significant positive coefficients, indicating wider spreads during weekends. In contrast, BTCFDUSD and ETHUSDT on Binance show significant negative coefficients, suggesting tighter spreads during weekends. These divergent patterns may reflect different types of market participants and trading behaviors across exchanges and cryptocurrency pairs during non-business days.

Finally, the explanatory power of our model varies across cryptocurrency pairs. The highest R^2 values are observed for ETH-USDT on Coinbase (0.28), BTCFDUSD on Binance (0.25), and ETHFDUSD on Binance (0.23), indicating that our model captures a substantial portion of the variation in spreads for these pairs. The lowest R^2 is for BTCUSDT on Binance (0.04), suggesting that additional factors may be driving liquidity for this pair. Overall, the model performs well for most cryptocurrency pairs, with a median R^2 of approximately 0.17 across all pairs.

Table 3 reports the estimates. Three patterns stand out. First, volatility is the strongest determinant of spreads. The EWMA measure σ_t loads positively and very significantly across all pairs, whereas the instantaneous proxy r_t^2 shows weaker and less uniform effects. This is exactly what inventory- and information-risk models predict when market makers price the cost of holding risk and trading against informed flow (Ho and Stoll, 1981; Huang and Stoll, 1997), and it accords with crypto evidence that realized volatility is the dominant driver of illiquidity and liquidity variation across venues (Ahmed, 2024; Brauneis et al., 2022b).

Second, coefficients on contemporaneous and lagged returns are typically negative, implying wider spreads when prices fall. This asymmetric liquidity response is consistent with dealers managing inventory and adverse-selection risk more conservatively in downturns (Hasbrouck, 1991) and with evidence that market liquidity deteriorates when markets decline (Hameed et al., 2010). The persistence in r_{t-1} echoes the idea that liquidity replenishment is gradual after adverse price moves.

Third, the periodicity factor (FFT_t) is often significant, though its sign varies by pair and

venue. This variation reflects differences in phase and session timing—i.e., whether the local peak of the daily cycle coincides with higher or lower spreads for a given market—and is in line with documented intraday periodicity in crypto centered on European–U.S. hours (Brauneis et al., 2025; Hansen et al., 2024b) and with exchange-specific microstructure heterogeneity in fragmented crypto markets (Makarov and Schoar, 2020).

Regarding calendar effects, weekend indicators are significant for roughly half of the pairs but with mixed signs. This heterogeneity matches prior findings that day-of-week regularities in crypto are weaker and less stable than the strong time-of-day cycle (Baur et al., 2019; Eross et al., 2019; Wang et al., 2020), and it likely reflects differences in participant mix across venues and quote currencies (Makarov and Schoar, 2020).

Finally, while several pairs attain sizeable R^2 (up to 0.28), explanatory power varies across markets. Modest R^2 for parsimonious intraday models are typical in high-frequency settings—periodic terms are highly informative but explain only part of the unconditional variation (Andersen and Bollerslev, 1997)—and our cross-pair results are consistent with commonality in liquidity interacting with venue-specific features (Hasbrouck, 2001).

In addition, to quantify the relative importance of each explanatory variable, we use a standardized approach that takes into account the different scales and distributions of the predictors. We standardize all explanatory variables and the dependent variable to have a mean of zero and a standard deviation of one, and fit an OLS regression model with the standardized variables. We then calculate the absolute value of each standardized coefficient and normalize these absolute values by dividing by their sum, expressing the contribution of each variable as a proportion of the total explained effect.

Figure 7 shows the relative importance of each explanatory variable across all cryptocurrency pairs. For all pairs, volatility (σ_t) emerges as the overwhelmingly dominant factor, accounting for 40-70% of the explained variation in spreads. Returns (both contemporaneous and lagged) collectively make a substantial contribution, particularly for pairs on Coinbase. The cyclical component and weekend effect show varying levels of importance

Table 3: Regression Results for Log-Spread

Symbol	α	Returns				Volatility		Periodicity	Calendar	R^2
		r_t	r_{t-1}	r_{t-2}	r_{t-3}	r_t^2	σ_t	FFT	Weekend	
Panel A: Binance										
BTCFDUSD	0.03 (4.48)	-0.01 (-1.85)	-0.03 (-4.70)	-0.02 (-4.40)	-0.01 (-2.28)	0.07 (5.24)	0.43 (45.04)	-0.07 (-10.53)	-0.12 (-7.42)	0.25
BTCUSDC	0.00 (0.40)	-0.03 (-5.00)	-0.02 (-3.79)	-0.01 (-2.68)	-0.01 (-2.07)	0.04 (3.46)	0.39 (40.49)	-0.04 (-4.87)	-0.01 (-0.72)	0.18
BTCUSDT	-0.00 (-0.45)	-0.03 (-4.56)	-0.02 (-4.03)	-0.01 (-1.15)	-0.01 (-1.29)	0.02 (2.64)	0.18 (24.86)	0.02 (3.06)	0.01 (0.81)	0.04
ETHFDUSD	-0.03 (-3.69)	-0.04 (-5.79)	-0.04 (-5.11)	-0.02 (-3.70)	-0.01 (-1.56)	0.09 (2.23)	0.44 (34.51)	0.00 (0.17)	0.10 (6.98)	0.23
ETHUSDC	0.00 (0.33)	-0.04 (-5.44)	-0.03 (-4.34)	-0.01 (-2.50)	-0.01 (-2.57)	0.06 (1.92)	0.28 (25.56)	-0.03 (-3.70)	-0.01 (-0.59)	0.11
ETHUSDT	0.02 (1.91)	-0.02 (-3.61)	-0.02 (-3.28)	-0.02 (-3.10)	0.00 (0.12)	0.02 (1.32)	0.26 (31.51)	-0.01 (-1.39)	-0.05 (-3.48)	0.08
Panel B: Coinbase										
BTC-EUR	-0.03 (-2.96)	-0.02 (-1.99)	-0.01 (-1.54)	-0.01 (-1.19)	-0.00 (-0.35)	0.13 (8.57)	0.33 (28.65)	-0.02 (-3.34)	0.09 (5.38)	0.16
BTC-USD	-0.06 (-6.45)	-0.03 (-5.42)	-0.03 (-4.63)	-0.02 (-4.16)	-0.02 (-3.34)	0.06 (7.46)	0.22 (28.97)	0.04 (6.03)	0.20 (12.28)	0.06
BTC-USDT	-0.04 (-4.97)	-0.03 (-3.67)	-0.02 (-3.60)	-0.02 (-2.53)	-0.01 (-1.92)	0.15 (9.91)	0.40 (40.65)	0.00 (0.48)	0.14 (8.99)	0.22
ETH-EUR	-0.00 (-0.49)	-0.03 (-2.77)	-0.03 (-3.79)	-0.03 (-3.32)	-0.02 (-2.70)	0.09 (3.33)	0.31 (22.75)	0.03 (3.79)	0.01 (0.80)	0.13
ETH-USD	-0.05 (-5.63)	-0.06 (-6.38)	-0.04 (-6.20)	-0.03 (-4.72)	-0.02 (-3.46)	0.06 (2.47)	0.39 (35.41)	0.01 (0.94)	0.16 (9.21)	0.17
ETH-USDT	-0.01 (-1.17)	-0.04 (-3.83)	-0.05 (-6.53)	-0.04 (-4.07)	-0.02 (-4.15)	0.09 (3.06)	0.47 (37.99)	-0.06 (-9.14)	0.03 (1.92)	0.28

Note: This table presents regression results for spread predictability, with variables grouped into three categories: (1) Returns, (2) Volatility, and (3) Intraday Periodicity. For each currency pair, the first row shows coefficient estimates and the second row (in parentheses) shows corresponding t-statistics. We apply a Bonferroni correction to our significance threshold to address the multiple testing problem arising from analyzing 12 different market pairs simultaneously. Bold and underlined values indicate coefficients with t-statistics having absolute values above 3 (we divide the standard 2.5% significance level by 18, resulting in an adjusted significance level of approximately 0.00139). Returns variables include contemporaneous returns (r_t) and three lags of returns (r_{t-1} through r_{t-3}). Volatility measures include squared returns (r_t^2) and exponentially weighted moving average volatility (σ_t). The periodicity component (FFT) captures cyclical intraday patterns identified through Fourier analysis. The R^2 column shows the model's explanatory power.

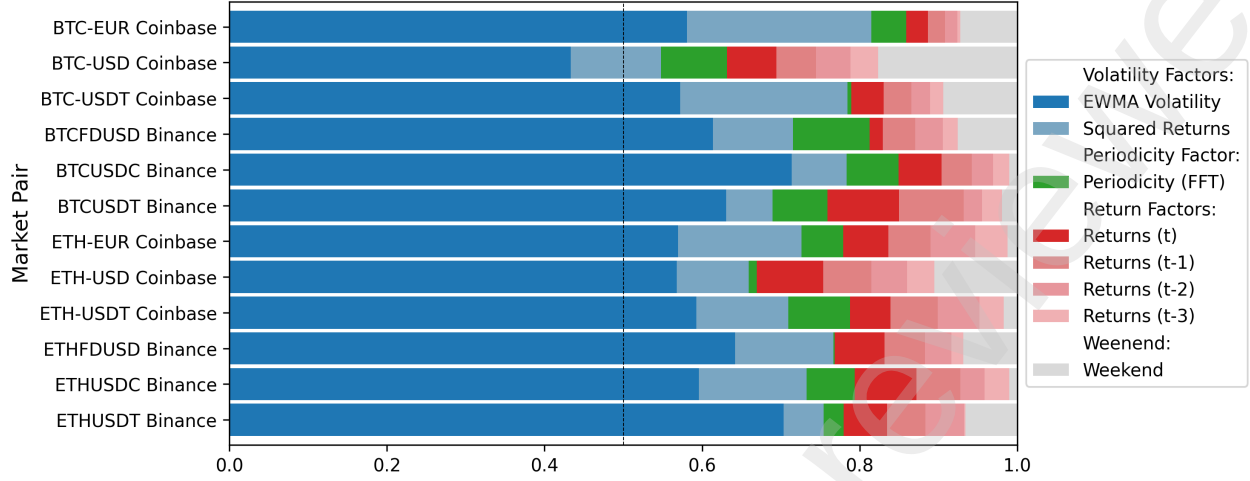
across different pairs, with the weekend effect being particularly important for BTC-USD and BTC-USDT on Coinbase.

6 Out-of-Sample Forecasting Performance

6.1 Forecasting Methodology

To assess the practical value of incorporating periodic patterns into liquidity modeling, we evaluate the out-of-sample predictive performance of our models. For our forecasting exer-

Figure 7: Relative Importance of Factors in Log-Spread Determination



Note: This table presents the normalized variable importance for each explanatory factor across all market pairs. Values represent the proportion of total explained effect attributable to each variable, calculated from the absolute standardized regression coefficients and normalized to sum to 1. Higher values indicate greater relative importance in explaining log-spread variation.

cise, we implement a more focused set of explanatory variables than in our full regression analysis, selecting variables with the strongest predictive power. We prepare a set of predictive features including:

- Lagged spread values (three lags of the spread variable: $\log(S_{t-1})$, $\log(S_{t-2})$, $\log(S_{t-3})$);
- Return measures (three lags: r_{t-1} , r_{t-2} , r_{t-3});
- Volatility measures (lagged volatility values: σ_{t-1} , σ_{t-2} , σ_{t-3});
- Mean reversion term (difference between the previous spread and its medium-term moving average - 96 previous observations);
- Intraday periodical component (sine wave with the dominant frequency and phase identified in our spectral analysis).

For each market pair, we split the data into training (60%) and testing (40%) sets. We estimate two model specifications on the training data: full model (including all variables with intraday periodical component) and restricted model (excluding intraday periodical component but retains all other variables). We generate out-of-sample forecasts for loga-

rithmic spreads using both models on the test data. We evaluate forecasting performance using multiple complementary metrics like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) nad Directional Accuracy (DA). We formally test for statistically significant differences in forecast accuracy using the Diebold-Mariano test, which evaluates whether one forecasting method consistently outperforms another.

Table 4: Out-of-Sample Forecast Performance Comparison

DM Test			Forecast-accuracy metrics									
Symbol	Stat	p-value	MSE		RMSE		MAE		MAPE (%)		DA (%)	
			Full	No FFT	Full	No FFT	Full	No FFT	Full	No FFT	Full	No FFT
Panel A: Binance												
BTCFDUSD	-1.605	0.108	0.081	0.081	0.285	0.285	0.222	0.223	6.00	6.00	33.10	33.00
BTCUSDC	-5.539	0.000	0.541	0.545	0.736	0.738	0.489	0.490	9.80	9.80	34.80	34.70
BTCUSDT	-0.375	0.708	7.729	7.731	2.780	2.781	2.563	2.564	36.00	36.00	40.60	40.00
ETHFDUSD	0.985	0.324	0.037	0.037	0.193	0.193	0.139	0.139	4.60	4.60	33.30	33.10
ETHUSDC	3.721	0.000	0.083	0.082	0.287	0.287	0.189	0.188	6.10	6.10	33.70	33.70
ETHUSDT	-0.401	0.689	1.028	1.028	1.014	1.014	0.789	0.788	14.90	14.90	34.80	34.80
Panel B: Coinbase												
BTC-EUR	-3.674	0.000	0.068	0.068	0.261	0.261	0.187	0.187	7.90	7.90	33.90	34.00
BTC-USD	-2.136	0.033	0.281	0.281	0.530	0.530	0.304	0.305	7.10	7.10	34.00	33.80
BTC-USDT	-2.473	0.013	0.039	0.039	0.198	0.198	0.140	0.141	5.60	5.60	33.60	33.60
ETH-EUR	-2.776	0.006	0.048	0.048	0.219	0.219	0.153	0.153	7.60	7.60	32.60	32.50
ETH-USD	0.939	0.348	0.049	0.049	0.222	0.222	0.151	0.151	4.80	4.80	33.50	33.50
ETH-USDT	-2.060	0.039	0.057	0.057	0.239	0.240	0.171	0.171	10.20	10.20	32.90	32.90

Note: This table presents forecast evaluation metrics comparing models with and without the periodicity (FFT) component. The Diebold-Mariano test evaluates whether forecast improvements are statistically significant, with negative values favoring the FFT model and bold underlined values indicating statistical significance at the 5% level or better. For each metric, the better performing model is highlighted in bold. For error metrics (RMSE, MAE, MAPE), lower values are better, while for directional accuracy (DA), higher values are better.

Table 4 presents the results of our forecasting analysis for cryptocurrency pairs across Binance and Coinbase exchanges. The Diebold-Mariano test results reveal that the inclusion of the FFT component significantly improves forecasting accuracy for several cryptocurrency pairs, though the pattern is not uniform across all pairs.

For Binance cryptocurrencies, BTCUSDC shows a highly significant improvement (DM = -5.539, p-value < 0.001) when incorporating the FFT component, suggesting that cyclical patterns substantially enhance predictive power for this pair. Interestingly, ETHUSDC shows a significant positive DM statistic (3.721, p-value < 0.001), indicating that for this particular

pair, the model without the FFT component performs better. The remaining four Binance pairs (BTCFDUSD, BTCUSDT, ETHFDUSD, and ETHUSDT) show DM statistics that are not statistically significant, suggesting that the inclusion of the FFT component neither significantly improves nor harms forecast accuracy for these pairs.

For Coinbase cryptocurrencies, four out of six pairs show statistically significant improvements with the inclusion of the FFT component: BTC-EUR (DM = -3.674, p-value < 0.001), BTC-USD (DM = -2.136, p-value = 0.033), BTC-USDT (DM = -2.473, p-value = 0.013), and ETH-EUR (DM = -2.776, p-value = 0.006). ETH-USDT also shows a significant improvement (DM = -2.060, p-value = 0.039). ETH-USD is the only Coinbase pair that does not show a significant difference between the models, with a positive but non-significant DM statistic (0.939, p-value = 0.348).

Examining the error metrics in detail, both MSE and RMSE generally favor the full model for most cryptocurrency pairs, though the improvements are often quite modest. For instance, BTCUSDC on Binance shows an MSE improvement from 0.545 to 0.541, while BTC-USDT on Coinbase shows an RMSE improvement from 0.198 to 0.198 (difference visible only at more decimal places). While these gains appear small numerically, their statistical significance in several cases (as confirmed by the Diebold-Mariano test) suggests consistent improvements that could yield economically meaningful benefits over numerous transactions.

Mean Absolute Error (MAE) follows a similar pattern to RMSE, with the full model generally outperforming the restricted model, albeit with small margins. In two cases (ETHUSDT on Binance and ETHUSDC on Binance), the restricted model actually performs marginally better in terms of MAE. Percentage errors (MAPE) show the full model consistently performing better across all cryptocurrency pairs, though the improvements are very small, typically around 0.1 percentage points or less.

In terms of directional accuracy (DA), which measures the ability to correctly predict the direction of spread changes, the full model generally performs better than the restricted model. BTC-USDT on Binance shows the largest improvement, with the full model correctly

predicting 40.6% of direction changes compared to 40.0% for the restricted model. For most other pairs, the improvements in directional accuracy are minimal (0.1-0.2 percentage points), and in one case (BTC-EUR on Coinbase), the restricted model actually achieves slightly higher directional accuracy (34.0% versus 33.9%).

The relatively modest improvements in forecast metrics despite statistically significant DM test results for several pairs suggest that while periodic patterns do contribute to forecast accuracy, their impact varies considerably across different cryptocurrency pairs and exchanges. This heterogeneity in forecasting performance may reflect differences in market microstructure, trading activity, or liquidity provision mechanisms across different cryptocurrency markets — consistent with evidence that expected returns to liquidity provision are amplified in smaller, more volatile, less liquid crypto pairs ([Bianchi et al., 2022](#)).

7 Concluding Remarks

This study provides an examination of market liquidity dynamics in the cryptocurrency markets, focusing on Bitcoin and Ethereum traded on two major exchanges: Binance and Coinbase. By analyzing high-frequency order book data through the lens of spectral analysis and statistical techniques, we have uncovered several key insights into the nature of liquidity provision in these relatively new but increasingly important markets.

Our results show that despite their 24/7 operation, cryptocurrency markets exhibit strong and statistically significant cyclical patterns in liquidity. These patterns are characterized by an inverted sinusoidal structure, with the lowest spreads typically observed during Asian and early European trading hours, followed by wider spreads during the North American trading session. The remarkable consistency of these patterns across cryptocurrencies and exchanges suggests the presence of global factors affecting the provision of liquidity in cryptocurrency markets, regardless of the specific trading venue or cryptocurrency.

The harmonic regression analysis shows that most cryptocurrency pairs share a domi-

nant frequency of approximately 0.01 cycles per observation (corresponding to a daily cycle), with additional secondary frequencies that likely reflect the influence of major global trading sessions. These periodic components are highly statistically significant, as evidenced by the robust t-statistics obtained in our regression analysis. This finding suggests that cryptocurrency liquidity is significantly influenced by traditional business hours and global trading patterns, despite the theoretical capacity of markets to operate around the clock.

Our analysis of weekend versus weekday trading patterns further supports this conclusion. We document a striking flattening of intraday spread patterns during weekends, with significantly lower volatility and generally tighter spreads compared to weekdays. This marked difference between weekend and weekday liquidity patterns strongly suggests that institutional trading activity, which follows traditional market hours, plays a key role in shaping the microstructure of the cryptocurrency market.

The integrated regression model shows that volatility is the dominant determinant of spread variation, accounting for 40-70% of the explained variation across all cryptocurrency pairs. This finding is consistent with risk-based theories of liquidity provision, where market makers demand wider spreads as compensation for taking positions during periods of higher price uncertainty. Returns also exhibit a significant and consistent negative relationship with spreads, suggesting that market liquidity tends to deteriorate during price declines. This asymmetric liquidity response to price movements may reflect risk management considerations by liquidity providers during market downturns.

The forecasting analysis shows that the inclusion of periodic components can improve the accuracy of spread predictions for many cryptocurrency pairs, particularly those traded on Coinbase. While the numerical improvements in predictive metrics are modest, the statistical significance of these improvements for several pairs suggests potential economic value for market participants engaged in high-frequency or high-volume trading.

Several avenues for future research emerge from our analysis. First, examining how these liquidity patterns have evolved over time, particularly as institutional participation in cryp-

tocurrency markets has increased, could provide insights into the evolution of the market. Second, exploring the relationship between liquidity patterns and the specific characteristics of different cryptocurrencies (such as market capitalization, use case, or technological features) could help identify factors that contribute to more robust liquidity. Third, examining how liquidity dynamics differ between centralized exchanges, decentralized exchanges, and over-the-counter markets would enhance our understanding of the cryptocurrency market ecosystem more broadly.

References

- Admati, A. R. and Pfleiderer, P. (1988). A Theory of Intraday Patterns: Volume and Price Variability. Review of Financial Studies, 1(1):3–40.
- Ahmed, W. M. A. (2024). On the robust drivers of cryptocurrency liquidity: The case of Bitcoin. Financial Innovation, 10(1):69.
- Amaya, D., Filbien, J.-Y., Okou, C., and Roch, A. F. (2018). Distilling liquidity costs from limit order books. Journal of Banking & Finance, 94:16–34.
- Amihud, Y. and Mendelson, H. (1980). Dealership market. Journal of Financial Economics, 8(1):31–53.
- Andersen, T. G. and Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. Journal of Empirical Finance, 4(2-3):115–158.
- Anderson, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2003). Micro Effects of Macro Announcements: Real-Time Price Discovery in Foreign Exchange. American Economic Review, 93(1):38–62.
- Baur, D. G., Cahill, D., Godfrey, K., and (Frank) Liu, Z. (2019). Bitcoin time-of-day, day-of-week and month-of-year effects in returns and trading volume. Finance Research Letters, 31:78–92.
- Benston, G. J. and Hagerman, R. L. (1974). Determinants of bid-asked spreads in the over-the-counter market. Journal of Financial Economics, 1(4):353–364.
- Bessembinder, H. (1994). Bid-ask spreads in the interbank foreign exchange markets. Journal of Financial Economics, 35(3):317–348.
- Bianchi, D. and Babiak, M. (2022). On the performance of cryptocurrency funds. Journal of Banking & Finance, 138:106467.

- Bianchi, D., Babiak, M., and Dickerson, A. (2022). Trading volume and liquidity provision in cryptocurrency markets. Journal of Banking & Finance, 142:106547.
- Black, J. R., Stock, D., and Yadav, P. K. (2016). The pricing of different dimensions of liquidity: Evidence from government guaranteed bonds. Journal of Banking & Finance, 71:119–132.
- Bouteska, A., Abedin, M. Z., Hajek, P., and Yuan, K. (2024). Cryptocurrency price forecasting – a comparative analysis of ensemble learning and deep learning methods. International Review of Financial Analysis, 92:103055.
- Brauneis, A., Mestel, R., Riordan, R., and Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? Journal of Banking & Finance, 124:106041.
- Brauneis, A., Mestel, R., Riordan, R., and Theissen, E. (2022a). The anatomy of a fee change — evidence from cryptocurrency markets. Journal of Empirical Finance, 67:152–167.
- Brauneis, A., Mestel, R., Riordan, R., and Theissen, E. (2022b). Bitcoin unchained: Determinants of cryptocurrency exchange liquidity. Journal of Empirical Finance, 69:106–122.
- Brauneis, A., Mestel, R., and Theissen, E. (2025). The crypto world trades at tea time: Intraday evidence from centralized exchanges across the globe. Review of Quantitative Finance and Accounting, 64(1):275–304.
- Cakici, N., Shahzad, S. J. H., Będowska-Sójka, B., and Zaremba, A. (2024). Machine learning and the cross-section of cryptocurrency returns. International Review of Financial Analysis, 94:103244.
- Cakici, N. and Zaremba, A. (2021). Liquidity and the cross-section of international stock returns. Journal of Banking & Finance, 127:106123.
- Cattivelli, L. and Pirino, D. (2019). A SHARP model of bid–ask spread forecasts. International Journal of Forecasting, 35(4):1211–1225.

- Cheng, J., Tiwari, S., Khaled, D., Mahendru, M., and Shahzad, U. (2024). Forecasting bitcoin prices using artificial intelligence: Combination of ml, sarima, and facebook prophet models. Technological Forecasting and Social Change, 198:122938.
- Chordia, T., Sarkar, A., and Subrahmanyam, A. (2005). An Empirical Analysis of Stock and Bond Market Liquidity. Review of Financial Studies, 18(1):85–129.
- Copeland, T. E. and Galai, D. (1983). Information Effects on the Bid-Ask Spread. The Journal of Finance, 38(5):1457.
- Corwin, S. A. and Schultz, P. (2012). A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. The Journal of Finance, 67(2):719–760.
- Dyhrberg, A. H., Foley, S., and Svec, J. (2018). How investible is Bitcoin? Analyzing the liquidity and transaction costs of Bitcoin markets. Economics Letters, 171:140–143.
- Easley, D. and O'Hara, M. (1987). Price, trade size, and information in securities markets. Journal of Financial Economics, 19(1):69–90.
- Eross, A., McGroarty, F., Urquhart, A., and Wolfe, S. (2019). The intraday dynamics of bitcoin. Research in International Business and Finance, 49:71–81.
- Evans, K. P. (2011). Intraday jumps and US macroeconomic news announcements. Journal of Banking & Finance, 35(10):2511–2527.
- Fiszeder, P., Małecka, M., and Molnár, P. (2024). Robust estimation of the range-based garch model: Forecasting volatility, value at risk and expected shortfall of cryptocurrencies. Economic Modelling, 141:106887.
- Fong, K. Y. L., Holden, C. W., and Trzcinka, C. A. (2017). What Are the Best Liquidity Proxies for Global Research?*. Review of Finance, 21(4):1355–1401.
- French, K. R. and Roll, R. (1986). Stock return variances. Journal of Financial Economics, 17(1):5–26.

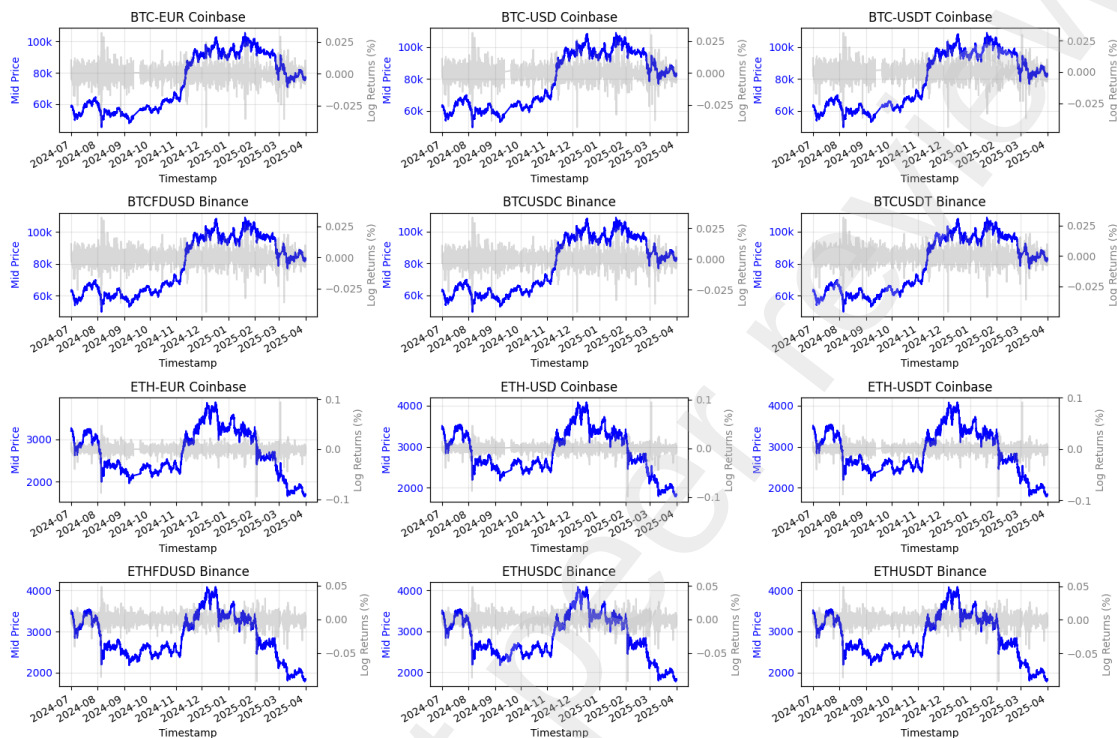
- Galati, L. (2024). Exchange market share, market makers, and murky behavior: The impact of no-fee trading on cryptocurrency market quality. Journal of Banking & Finance, 165:107222.
- Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics, 14(1):71–100.
- Griffith, T. G. and Roseman, B. S. (2019). Making cents of tick sizes: The effect of the 2016 U.S. SEC tick size pilot on limit order book liquidity. Journal of Banking & Finance, 101:104–121.
- Hameed, A., Kang, W., and Viswanathan, S. (2010). Stock Market Declines and Liquidity. The Journal of Finance, 65(1):257–293.
- Han, S. (2023). Is liquidity risk priced in cryptocurrency markets? Applied Economics Letters, 30(17):2481–2487.
- Hansen, P. R., Kim, C., and Kimbrough, W. (2024a). Periodicity in Cryptocurrency Volatility and Liquidity. Journal of Financial Econometrics, 22(1):224–251.
- Hansen, P. R., Kim, C., and Kimbrough, W. (2024b). Periodicity in Cryptocurrency Volatility and Liquidity. Journal of Financial Econometrics, 22(1):224–251.
- Hasbrouck, J. (1991). Measuring the Information Content of Stock Trades. The Journal of Finance, 46(1):179–207.
- Hasbrouck, J. (2001). Common factors in prices, order flows, and liquidity. Journal of Financial Economics, 59(3):383–411.
- Ho, T. and Stoll, H. R. (1981). Optimal dealer pricing under transactions and return uncertainty. Journal of Financial Economics, 9(1):47–73.

- Huang, R. D. and Stoll, H. R. (1997). The Components of the Bid-Ask Spread: A General Approach. Review of Financial Studies, 10(4):995–1034.
- Jahanshahloo, H., Corbet, S., and Oxley, L. (2022). Seeking sigma: Time-of-the-day effects on the Bitcoin network. Finance Research Letters, 49:103101.
- Kyle, A. S. (1985). Continuous Auctions and Insider Trading. Econometrica, 53(6):1315.
- Maciel, L., Ballini, R., Gomide, F., and Yager, R. (2022). Forecasting cryptocurrencies prices using data driven level set fuzzy models. Expert Systems with Applications, 210:118387.
- Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. Journal of Financial Economics, 135(2):293–319.
- McInish, T. H. and Wood, R. A. (1990). An analysis of transactions data for the Toronto Stock Exchange. Journal of Banking & Finance, 14(2-3):441–458.
- McInish, T. H. and Wood, R. A. (1992). An Analysis of Intraday Patterns in Bid/Ask Spreads for NYSE Stocks. The Journal of Finance, 47(2):753–764.
- Nasirtafreshi, I. (2022). Forecasting cryptocurrency prices using recurrent neural network and long short-term memory. Data and Knowledge Engineering, 139:102009.
- Rakowski, D. and Wang Beardsley, X. (2008). Decomposing liquidity along the limit order book. Journal of Banking & Finance, 32(8):1687–1698.
- Rösch, C. G. and Kaserer, C. (2013). Market liquidity in the financial crisis: The role of liquidity commonality and flight-to-quality. Journal of Banking & Finance, 37(7):2284–2302.
- Scharnowski, S. (2021). Understanding Bitcoin liquidity. Finance Research Letters, 38:101477.

- Stoll, H. R. (1989). Inferring the Components of the Bid-Ask Spread: Theory and Empirical Tests. The Journal of Finance, 44(1):115–134.
- Tzeng, K.-Y. and Su, Y.-K. (2024). Can u.s. macroeconomic indicators forecast cryptocurrency volatility? The North American Journal of Economics and Finance, 74:102224.
- Wang, J.-N., Liu, H.-C., and Hsu, Y.-T. (2020). Time-of-day periodicities of trading volume and volatility in Bitcoin exchange: Does the stock market matter? Finance Research Letters, 34:101243.
- Wei, W. C. (2018). Liquidity and market efficiency in cryptocurrencies. Economics Letters, 168:21–24.
- Zaremba, A., Bilgin, M. H., Long, H., Mercik, A., and Szczygielski, J. J. (2021). Up or down? short-term reversal, momentum, and liquidity effects in cryptocurrency markets. International Review of Financial Analysis, 78:101908.
- Zhang, W. and Li, Y. (2023). Liquidity risk and expected cryptocurrency returns. International Journal of Finance & Economics, 28(1):472–492.

A Appendix

Figure A.1: Time series of mid prices for cryptocurrency and traditional currency pairs from July 1, 2024 to February 16, 2025.



Note: The chart shows the evolution of mid prices calculated as $(P_{ask}^t + P_{bid}^t)/2$ for the analyzed market pairs. Cryptocurrency prices display higher volatility compared to the traditional currency pairs. All series are normalized to their initial values for comparison purposes.

Table A.1: Stationarity Test Results for Cryptocurrency Market Spreads

Symbol	ADF		DFGLS		PP		ZA		KPSS	
	Stat	p-val	Stat	p-val	Stat	p-val	Stat	p-val	Stat	p-val
Symbol										
BTC-EUR Coinbase	-12.47	0.0	-11.09	0.0	-140.63	0.0	-13.96	0.0	1.20	0.01
BTC-USD Coinbase	-8.78	0.0	-6.45	0.0	-155.73	0.0	-11.24	0.0	1.31	0.01
BTC-USDT Coinbase	-16.93	0.0	-10.71	0.0	-80.78	0.0	-17.47	0.0	0.55	0.01
BTCFDUSD Binance	-11.71	0.0	-11.59	0.0	-182.80	0.0	-12.29	0.0	0.52	0.01
BTCUSDC Binance	-8.16	0.0	-8.00	0.0	-154.23	0.0	-12.17	0.0	1.30	0.01
BTCUSDT Binance	-14.61	0.0	-6.92	0.0	-196.81	0.0	-15.43	0.0	0.83	0.01
ETH-EUR Coinbase	-16.09	0.0	-13.06	0.0	-83.99	0.0	-17.81	0.0	0.56	0.01
ETH-USD Coinbase	-13.68	0.0	-6.69	0.0	-100.87	0.0	-14.71	0.0	0.93	0.01
ETH-USDT Coinbase	-15.74	0.0	-12.87	0.0	-132.75	0.0	-16.55	0.0	0.53	0.01
ETHFDUSD Binance	-11.32	0.0	-5.17	0.0	-169.32	0.0	-11.88	0.0	0.34	0.01
ETHUSDC Binance	-7.50	0.0	-4.98	0.0	-140.26	0.0	-10.56	0.0	2.32	0.01
ETHUSDT Binance	-11.03	0.0	-4.20	0.0	-199.37	0.0	-11.34	0.0	0.41	0.01

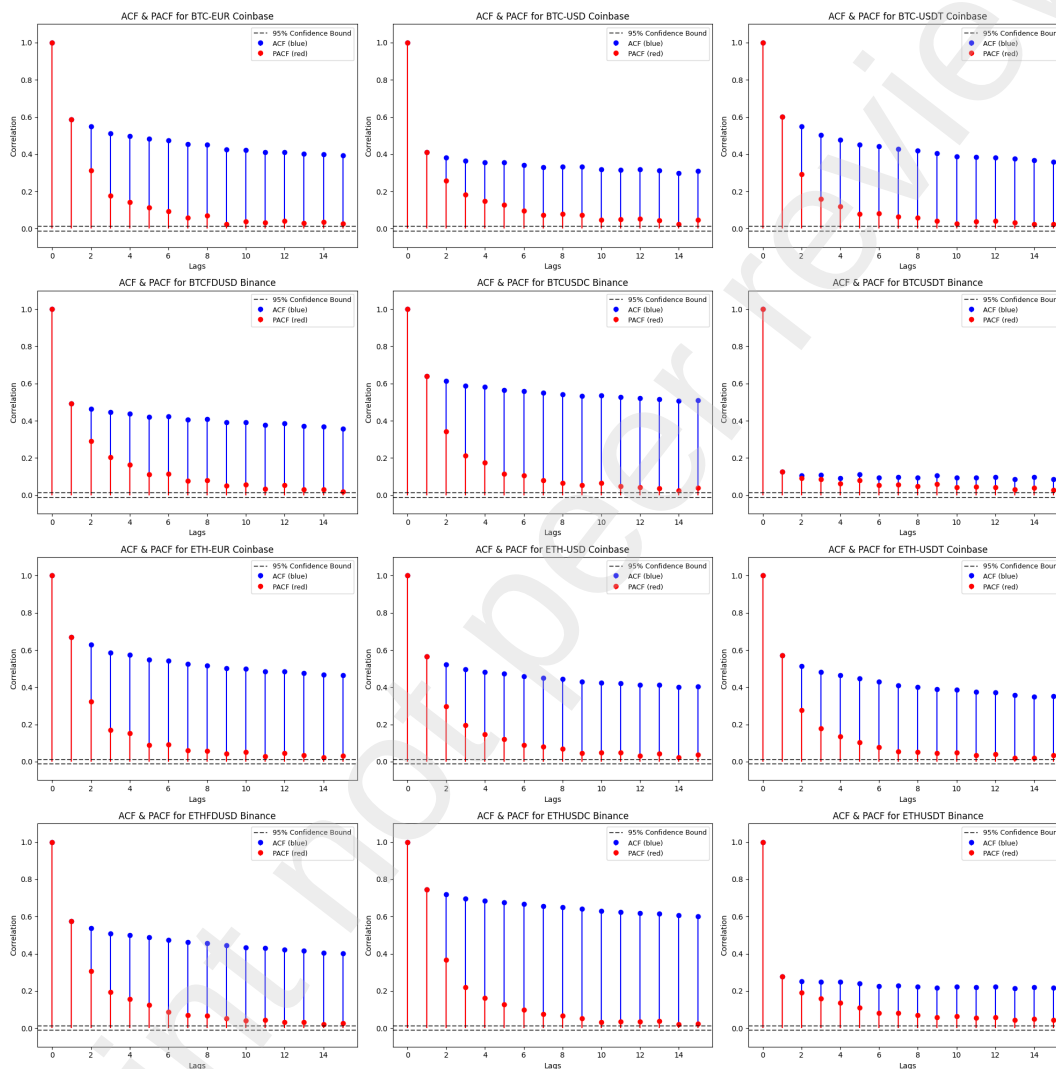
Note: The table presents the results of stationarity tests for the spread series of 12 cryptocurrency markets. Both the Augmented Dickey-Fuller (ADF) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests were run. The ADF test consistently rejects the null hypothesis of a unit root for all series, while the KPSS test gives mixed results, with some markets showing signs of non-stationarity. These tests were performed on spreads calculated at 15-minute intervals, collected from 1 July 2024 to 31 March 2025.

Table A.2: Results of the Box-Pierce and Ljung-Box Tests for Autocorrelation in Spread Series

Symbol	Box-Pierce Stat	Box-Pierce p-value	Ljung-Box Stat	Ljung-Box p-value
BTC-EUR Coinbase	20427.45	0.0	20431.93	0.0
BTC-USD Coinbase	40414.08	0.0	40424.82	0.0
BTC-USDT Coinbase	22796.32	0.0	22800.14	0.0
BTCFDUSD Binance	45298.06	0.0	45310.39	0.0
BTCUSDC Binance	71069.75	0.0	71089.14	0.0
BTCUSDT Binance	4889.24	0.0	4890.56	0.0
ETH-EUR Coinbase	23845.75	0.0	23849.67	0.0
ETH-USD Coinbase	26605.78	0.0	26610.90	0.0
ETH-USDT Coinbase	8759.59	0.0	8761.13	0.0
ETHFDUSD Binance	47721.17	0.0	47733.93	0.0
ETHUSDC Binance	88257.49	0.0	88281.88	0.0
ETHUSDT Binance	10472.65	0.0	10475.42	0.0

Note: This table presents the test statistics and p-values from the Box-Pierce and Ljung-Box tests applied to the spread series of various cryptocurrency pairs (symbols). Both tests assess the presence of autocorrelation in the spread data up to a specified number of lags. These tests were performed on spreads calculated at 15-minute intervals, collected from 1 July 2024 to 31 March 2025.

Figure A.2: Partial Autocorrelation (PACF) Plots for Cryptocurrency Market Spreads



Note: The table shows the results of autocorrelation analysis for the spread series of 12 cryptocurrency markets using the Box-Pierce and Ljung-Box tests. Both tests indicate significant autocorrelation in the spread series, with p-values of 0.00 across all markets. This highlights the persistence of spread values over time and suggests predictive patterns within liquidity dynamics. These tests were performed on spreads calculated at 15-minute intervals, collected from from 1 July 2024 to 31 March 2025.

Figure A.3: Day-of-Week Analysis of Spread Patterns for Cryptocurrency Pairs

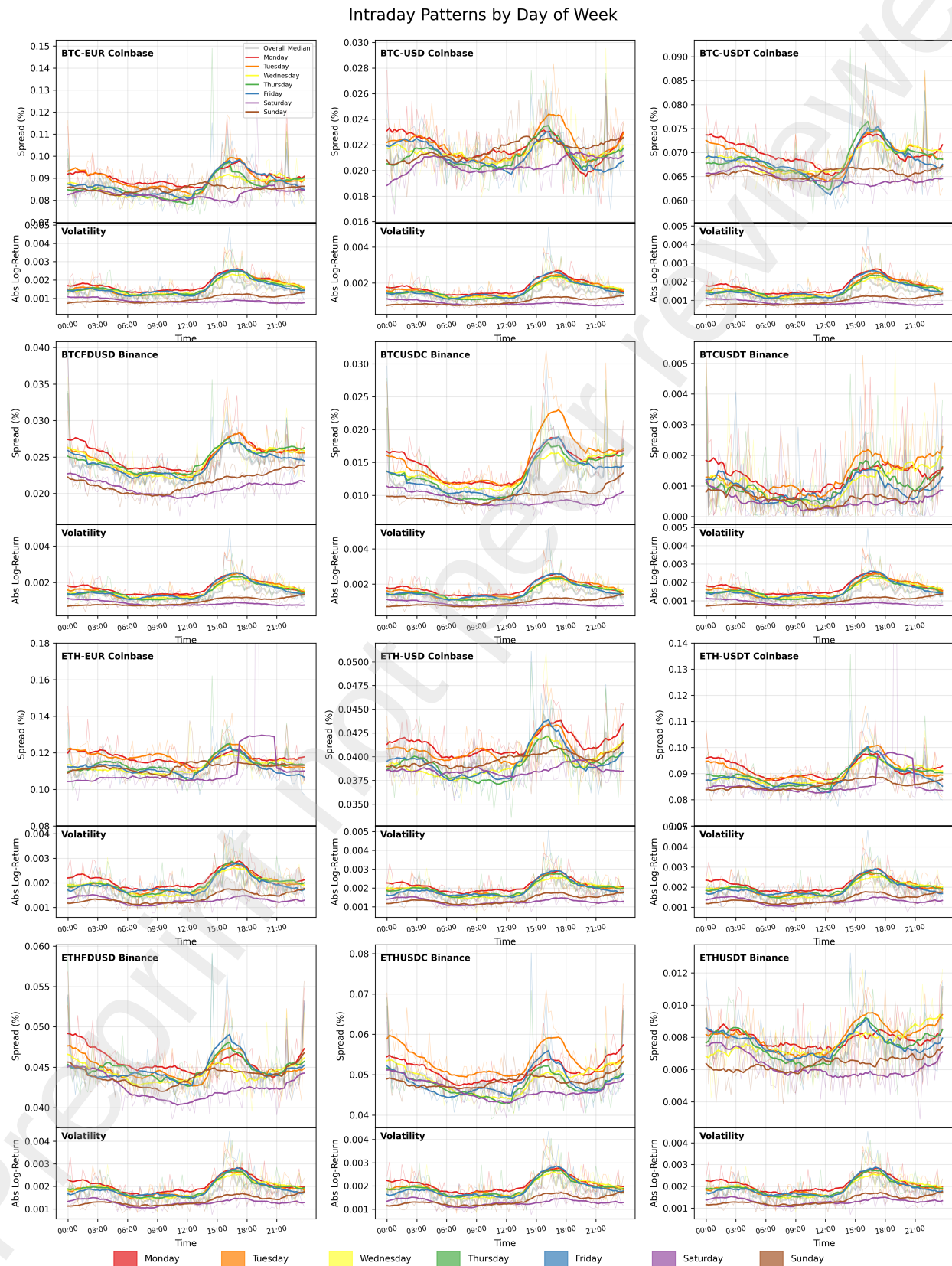


Table A.3: Stationarity Test Results for Cryptocurrency Market Log-Spreads

Symbol	ADF		DFGLS		PP		ZA		KPSS	
	Stat	p-val	Stat	p-val	Stat	p-val	Stat	p-val	Stat	p-val
BTC-EUR Coinbase	-7.66	0.00	-6.24	0.00	-164.22	0.0	-12.16	0.00	0.94	0.01
BTC-USD Coinbase	-12.15	0.00	-9.46	0.00	-194.13	0.0	-13.30	0.00	0.12	0.09
BTC-USDT Coinbase	-9.51	0.00	-7.36	0.00	-181.48	0.0	-11.25	0.00	1.06	0.01
BTCFDUSD Binance	-12.10	0.00	-10.78	0.00	-198.12	0.0	-12.72	0.00	0.24	0.01
BTCUSDC Binance	-9.84	0.00	-9.84	0.00	-210.24	0.0	-12.12	0.00	1.00	0.01
BTCUSDT Binance	-8.46	0.00	-4.68	0.00	-212.85	0.0	-15.85	0.00	0.56	0.01
ETH-EUR Coinbase	-8.49	0.00	-8.49	0.00	-164.50	0.0	-11.90	0.00	0.75	0.01
ETH-USD Coinbase	-9.64	0.00	-3.02	0.00	-195.71	0.0	-11.06	0.00	1.12	0.01
ETH-USDT Coinbase	-10.44	0.00	-10.43	0.00	-176.39	0.0	-11.33	0.00	0.46	0.01
ETHFDUSD Binance	-9.55	0.00	-7.93	0.00	-200.07	0.0	-12.17	0.00	0.87	0.01
ETHUSDC Binance	-5.94	0.00	-4.57	0.00	-200.36	0.0	-8.07	0.00	3.45	0.01
ETHUSDT Binance	-8.09	0.00	-7.87	0.00	-224.38	0.0	-10.91	0.00	1.59	0.01

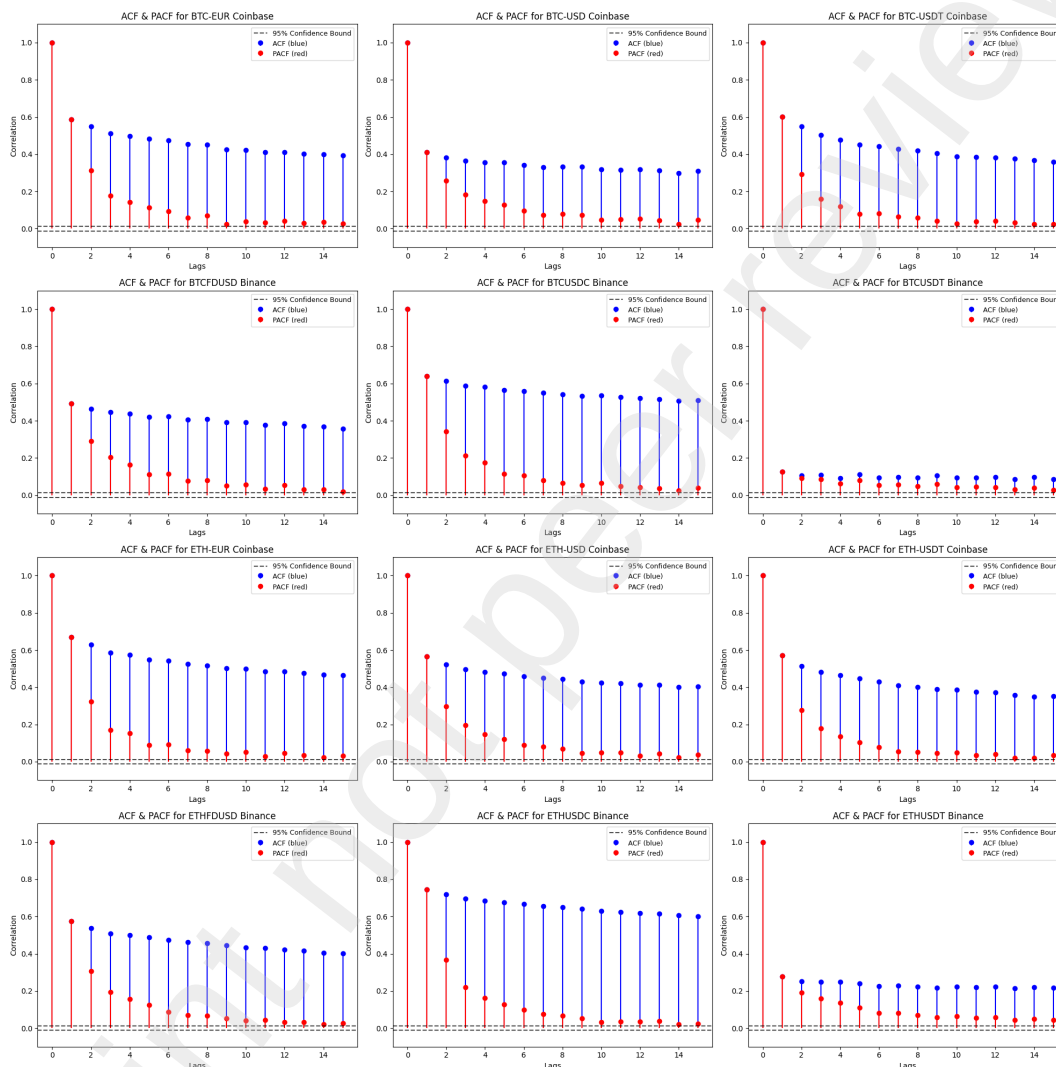
Note: The table presents the results of stationarity tests for the logarithms of the spread series of 12 cryptocurrency markets. We consider the Augmented Dickey-Fuller (ADF), DF-GLS, Phillips-Perron (PP), Zivot-Andrews (ZA) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests. These tests were performed on spreads calculated at 15-minute intervals, collected from 1 July 2024 to 31 March 2025.

Table A.4: Results of the Box-Pierce and Ljung-Box Tests for Autocorrelation in Spread Series

Symbol	Box-Pierce Stat	Box-Pierce p-value	Ljung-Box Stat	Ljung-Box p-value
BTC-EUR Coinbase	70065.93	0.0	70086.17	0.0
BTC-USD Coinbase	9773.53	0.0	9776.21	0.0
BTC-USDT Coinbase	36909.35	0.0	36919.56	0.0
BTCFDUSD Binance	33251.93	0.0	33261.11	0.0
BTCUSDC Binance	30663.87	0.0	30672.48	0.0
BTCUSDT Binance	36945.82	0.0	36956.38	0.0
ETH-EUR Coinbase	56426.98	0.0	56442.87	0.0
ETH-USD Coinbase	27926.11	0.0	27933.98	0.0
ETH-USDT Coinbase	37809.88	0.0	37820.30	0.0
ETHFDUSD Binance	32144.03	0.0	32152.84	0.0
ETHUSDC Binance	55566.84	0.0	55582.72	0.0
ETHUSDT Binance	22205.16	0.0	22211.49	0.0

Note: This table presents the test statistics and p-values from the Box-Pierce and Ljung-Box tests applied to the spread series of various cryptocurrency pairs (symbols). Both tests assess the presence of autocorrelation in the spread data up to a specified number of lags. These tests were performed on spreads calculated at 15-minute intervals, collected from 1 July 2024 to 31 March 2025.

Figure A.4: Partial Autocorrelation (PACF) Plots for Cryptocurrency Market Spreads



Note: The table shows the results of autocorrelation analysis for the spread series of 12 cryptocurrency markets using the Box-Pierce and Ljung-Box tests. Both tests indicate significant autocorrelation in the spread series, with p-values of 0.00 across all markets. This highlights the persistence of spread values over time and suggests predictive patterns within liquidity dynamics. These tests were performed on spreads calculated at 15-minute intervals, collected from from 1 July 2024 to 31 March 2025.